Learning to reduce the cost of Feynman integral reduction with Kira

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Abstract

Integration-by-parts reduction expresses Feynman integrals in a master basis [1]. We combine exact formulas, independent-loop factorization, basis-compatible native settings, and a trained sparse-seed expert. An outer model learns complete route costs, including prediction, equation generation, algebra, exact checks, and recovery. Separate validation compares the resulting method with the selected ML-free control before enabling learned decisions. The hybrid test contains 34 requests, of which 34 pass three exact checks. For 24 matched requests, the integrated method takes 136.956 seconds and the validation-selected ML-free control takes 143.079 seconds. This gives 4.3% less time, with ratio 0.957 and repetition interval $[0.925, 0.998]$. The comparison with the selected ML-free control resolves an added saving from learned decisions beyond the measured repetition variation. On the 8 requests with index vectors absent from outer-model training, the method takes 9.9% more time than the preceding application. A separate 65-request baseline test covers specified analytic families and independent products through thirteen loops, while coupled native calculations reach three loops. The sparse-expert comparison uses a nonplanar two-loop family with numerator ranks two through four. We keep the dimension symbolic and use rational values for other invariants. These timings assume a prepared master basis; training and independent reference preparation remain separate costs. General high-loop scattering and globally minimum reduction paths remain unverified.

1 Introduction

Feynman integrals arise when we integrate over the internal loop momenta in higher-order perturbative corrections to scattering amplitudes and particle decay amplitudes. In quantum chromodynamics, these corrections contribute to predictions for cross sections and decay rates. Many integrals in the same calculation have related numerator and denominator structures, and integration-by-parts (IBP) identities allow us to express them in terms of a smaller set of master integrals [2, 3].

Kira automates this reduction by generating identities and solving the resulting linear system [1, 4]. Its existing seed and equation-selection methods already avoid part of the unnecessary work [5]. Our question is whether a model can make an earlier choice that reduces the remaining calculation time. We study seed selection, the choice of an elimination row, and the selection of individual equations. Throughout this paper, a target is an integral or linear combination requested by the user; a seed is an index vector at which Kira generates identities.

Previous studies have used machine-assisted search to improve seed regions, including sparse regions around target-dependent paths [6, 7]. Our comparisons use the time required to complete an exact reduction, including prediction and unsuccessful attempts. This criterion follows directly from the purpose of the model: a smaller equation set is useful only when its reduction and recovery cost give a smaller total time. Section 3 defines the corresponding controls.

We first identify the measured benefit of the three local decisions and retain the seed and equation models where their tests support their use. Second, we train a shared cost model from

structural features and test its transfer to families and loop orders omitted from training. A further comparison trains on failed attempts and includes the preparation workflow. We then extend the application through thirteen loops for specified factorable and analytic families, and retrain the final selector on complete timings with the current master-basis checks. We first define the notation used in this paper, then explain the models and compare their results. The companion notes preserve historical variants, secondary experiments, and longer derivations.

2 The reduction problem and the three decisions

An integral family specifies inverse propagators and external kinematics. We use the convention [2, 1]

$$I_{\boldsymbol{a}}(\boldsymbol{x}, d) = \int \prod_{h=1}^{L} \frac{d^d \ell_h}{i\pi^{d/2}} \prod_{j=1}^{N} D_j^{-a_j}, \qquad D_j = q_j^2 - m_j^2 + i0. \tag{1}$$

Here L is the number of loops, ℓ_h is loop momentum h, q_j is a fixed linear combination of loop and external momenta, and m_j is the mass associated with propagator j. The vector $\boldsymbol{x}$ contains squared masses and external momentum invariants. Dimensional regularization uses $d = 4 - 2\epsilon$, where ϵ is the regulator, and $i0$ specifies the Feynman pole prescription. The normalization of the measure is a convention. A positive integer a_j places D_j in the denominator, a negative integer places it in the numerator, and zero removes that factor. The list of N entries includes auxiliary inverse propagators when the physical denominators alone do not span the loop scalar products [2].

Integrals with the same positive-index positions belong to the same sector. A subsector removes some of those positions [1]. At seed $\boldsymbol{\sigma}$, differentiating the integrand with respect to loop momentum ℓ_k gives the identity [3]

$$0 = \int \prod_{h} \frac{d^d \ell_h}{i\pi^{d/2}} \frac{\partial}{\partial \ell_k^\mu} \left(v^\mu \prod_j D_j^{-\sigma_j} \right) = \sum_u A_{\boldsymbol{\sigma},o;u}(\boldsymbol{x}, d) I_{\boldsymbol{a}^{(u)}}. \tag{2}$$

The vector v is a loop or external momentum, and o labels the pair (k, v). The coefficient $A_{\boldsymbol{\sigma},o;u}$ multiplies the integral with index vector $\boldsymbol{a}^{(u)}$. Appendix A expands the derivative and derives the resulting index shifts. Several operators act on the same seed, so seed selection retains or removes a group of candidate equations.

Kira combines IBP identities with symmetry and Lorentz-invariance relations, removes redundant equations, and eliminates integrals in a chosen order. Back-substitution then expresses the requested targets in the remaining master basis [1, 5]. For target T, the result has the form

$$T = \sum_{m=1}^{n_M} r_{Tm}(\boldsymbol{x}, d) M_m, \qquad r_{Tm} \in \mathbb{Q}(\boldsymbol{x}, d). \tag{3}$$

Here M_m is master integral m, n_M is the number of masters, and $\mathbb{Q}(\boldsymbol{x}, d)$ is the field of rational functions with rational coefficients. Equation (3) follows by the exact row operations derived in Appendix A. It determines reduction coefficients; evaluating the masters is a further part of an amplitude calculation [2]. Kira can use symbolic rational arithmetic or repeated finite-field solutions followed by rational reconstruction [8]. We leave reconstruction unchanged in this study.

Table 1 locates the three learned decisions. Our implementation always obtains the coefficients through exact algebra. If a restricted system does not reach the supplied basis, it repeats the reduction with less restrictive settings, and the timer includes this recovery. Exactness follows from the retained identities and row operations; sufficiency must still be checked for each target, as explained in Appendix A.

Table 1: The three places where a learned decision can change the work. These are the interventions tested in this study.

Decision	Model output	Work affected
Seed selection	Seed region or seed-retention score	Number and structure of identities generated
Row selection	Score for each allowed source row	Intermediate expressions during elimination
Equation selection	Retention score for a seed–operator pair	Symbolic coefficient construction

3 Training and comparison design

We use two kinds of labels. Early models learn membership in a recorded sufficient seed set or the next action in a recorded reduction. Such labels describe one valid history and do not establish a minimum number of seeds or operations. Later cost models learn from complete Kira timings for alternative choices. Section 6 uses this second objective to test whether one model can transfer across structures. The notes retain the reverse-construction dataset and its separate label-accuracy tests.

Every reported paired timing ratio uses identical targets, kinematics, and master bases for its competing methods. The native control disables learned choices in the same exact implementation. A fixed control uses a setting selected before testing and makes no request-dependent prediction. This second control measures what the prediction adds after fixed tuning. The earlier deployed application supplies a further practical comparison. When a study uses a modified executable, "native" means its native decision rules; it does not mean an unmodified Kira installation.

The main native comparisons keep d symbolic and assign rational values to the other kinematic variables. Thus the exported coefficients are exact rational functions of d. Master discovery and an enlarged-region reference calculation precede the supplied-basis timer. The application receives the master indices, while reference coefficients enter only the final check. We report this preparation cost separately where measured. Adding it to a supplied-basis time describes a two-stage workflow, since the reference calculation already performs a reduction; it does not measure a faster first reduction from scratch.

The new comparison starts a fresh process for each measurement and includes imports, model loading, prediction, IBP generation, elimination, exact coefficient comparison, retries, output, and process exit. Three repetitions rotate method order on the same worker for each request. Earlier retained comparisons use their stated timers and repetitions in Table 2. For the seed, equation, and shared-model workloads, we sum the median time of each request and divide by the corresponding control total. The historical row comparison uses a geometric mean, as stated in Section 4. Appendix E defines this ratio and its resampling interval. A nontrivial batch contains at least one target that reduces to a nonzero expression beyond an unchanged basis element. Failures remain in the coverage counts and receive no completed-time ratio.

4 Seed selection and elimination-row selection

4.1 Learning a sparse seed region

The sparse-seed model limits the distance between a seed's numerator powers and the numerator-power regions of the targets. Its boundary also depends on the seed rank and sector. We fit the boundary by running complete reductions, then train cost predictors to choose among candidate boundaries. Appendix B defines the geometry and objective. Trees and forests provide the

candidate cost models [9, 10, 11]; validation determines whether request-dependent prediction improves on a fixed sparse setting.

This retained comparison uses planar and nonplanar two-loop boxes with numerator ranks two through four, unit positive physical indices, rational kinematics, and symbolic d. Twelve training requests and ten validation requests per graph precede the reserved test. Validation retains a fixed boundary for the planar graph and enables a depth-two, 32-tree extremely randomized forest for the nonplanar graph. The 20 reserved requests have seven repetitions per method. All 560 timed reductions pass exact checks, with 5,600 coefficient comparisons. Table 2 shows 22.2% less time than tuned native choices and 3.6% less time than the fixed sparse control. These measurements test new parameter profiles within known graphs; they motivate, but do not establish, cross-family transfer.

4.2 Why the final method retains native row choices

The row model scores the allowed source rows of the current coefficient matrix. Its original network maps 132 inputs to 32 features, pools the available actions, and uses a 64-component hidden layer to produce a probability for each action. The inputs describe the pivot, nonzero entries, matrix size, and sampled cancellations. Exact arithmetic performs the selected row operation, as derived in Appendix A. The network learns the recorded next action through categorical cross-entropy; the notes give its full specification and the derivation of that loss.

In our live Kira comparison, the original row network takes 1.305 times the stock reduction time across 84 exactly checked trials. A later compiled policy trained on algebraic cost gives a geometric mean time ratio of 0.982 with repeat interval $[0.959, 1.003]$. Since this interval includes equal time, that experiment supplies no reliable complete-time gain. We therefore omit learned row selection from the retained method and leave Kira's row choices active. The historical row measurements and their different controls remain in the notes.

5 Selecting equations before coefficient construction

Equation selection makes a finer choice within the seed region. A classification tree scores each seed–operator pair before Kira builds its symbolic coefficients. Its seventeen features describe loop and external-momentum counts, index counts, target number, seed powers, sector membership, distance to a target, and operator index. The score is the weighted positive fraction in the reached leaf, using the tree construction of Ref. [9]. Appendix B.1 defines the features, weights, and threshold rule.

A positive training label means that the equation's set of participating integrals appears in Kira's selected system. This proxy does not identify a unique necessary equation. We collect 137,015 sampled records, including 39,285 positive labels, from 66 training batches. Shared and graph-specific trees use depths 6, 10, 14, or 18 and minimum leaf populations 4 or 16. Separate validation first retains high-recall candidates, then measures complete reduction cost. A graph uses a filter only when validation gives at least 5% less time than the same tuned calculation with all ML disabled. The selected filter can retain an individual equation or all operators from a retained seed.

The reserved comparison contains 68 matched nontrivial batches from a 74-request grid at one through four loops. Four reference checks fail and two batches are identity-only. Filtering reduces summed median time from 153.937 to 146.586 seconds, or 4.8%, with ratio interval $[0.946, 0.959]$. On the 32 batches where validation enables filtering, time falls from 91.707 to 84.384 seconds, or 8.0%. All 700 timed completions, including the identity-only batches, pass exact checks. Prediction and five recovery attempts remain inside the timer. These results use known graph structures and partly repeated target patterns.

A separate test changes the index batches within the enabled structures while keeping the model fixed. It retains only batches absent from training and validation, with at least one previously unseen index vector. All 24 requests complete, with 144 exact timed reductions and 5,106 coefficient comparisons. Their summed median time changes from 72.295 to 67.478 seconds, or 6.7%, with ratio interval $[0.928, 0.940]$. This index-transfer result supplies the equation-model entry in Table 2; it remains separate from transfer to new graphs.

Table 2: Direct results for the retained local decisions. Times are sums of request medians in seconds. Sparse seeding includes seven repetitions; equation filtering on new index batches includes three. Their workloads differ, so the ratios cannot be multiplied. These are measurements of this study.

Comparison	Control	Learned	Ratio
Sparse seeds / tuned native, 20 two-loop requests	75.843	58.972	0.778
Sparse seeds / fixed sparse, same requests	61.203	58.972	0.964
Equation filter / ML off, 24 new index batches	72.295	67.478	0.933

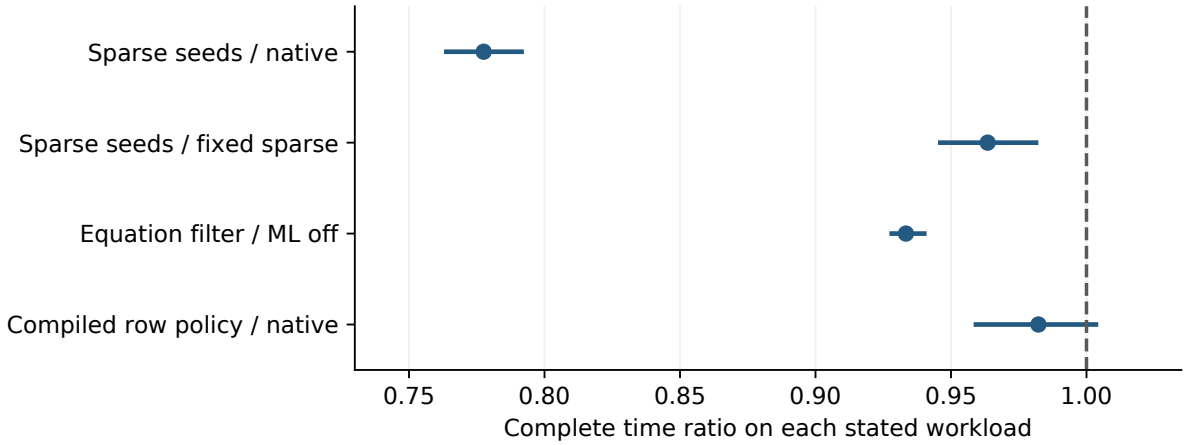


Figure 1: The retained local models reduce time on their stated workloads. Seed selection is compared with both native and fixed sparse choices. Equation selection changes target indices within known structures. Horizontal intervals describe paired repetition variation; the line at one marks equal time. Row selection remains omitted because its later interval includes one.

6 A shared model for transfer across integral families

The preceding gains depend on validated graph domains. We therefore train a shared model from structural quantities, with graph names, stored family signatures, and coordinate-position identifiers excluded from its inputs. Its 44 inputs summarize the family and targets through loop and external-momentum counts, propagator incidence, masses, positive and negative powers, sector occupancy, and normalized versions of these counts. These features have fixed length when the number of loops or propagators changes. Appendix C specifies the representation and prediction equations.

For each training request, the model predicts the complete cost of ten seed configurations, including changed positive-power bounds, numerator-rank bounds, and Kira's subsector truncation. We fit logarithms of time ratios to native bounds and penalize incomplete attempts. These labels include recovery to less restrictive configurations. A decision margin requires a predicted improvement before changing a common anchor action. The anchor and margin are global; no graph-specific deployment rule enters this new model.

We reuse only recorded training costs and reserve the earlier validation costs for model selection. The distinct-mass vacuum family of Appendix D, nonplanar boxes, pentagons, and loop orders four and above are excluded from both sets. The remaining data contain 74 training and 51 validation requests. Candidate models include nine tree or forest configurations, two regularized linear regressions, and two neural networks. Five-fold validation keeps each ordered graph structure within one fold, and the model selection also considers the separate validation set. Training selects a neural network with two 48-component hidden layers and a margin of 0.05. The network has 5,002 fitted weights and biases. The runtime evaluates the saved weights directly, and its predictions agree with those of the training library within the numerical check tolerance. Appendix C gives the loss, candidate settings, and selection rule.

The reserved physical profile is new. Its 110 requests span the defined one-mass and massive vacuum families, massless and massive two-point families, four-point ladders, boxes with bubble subgraphs, and three additional scattering structures. The declared loop range is one through twelve. This tests both new parameter values in training domains and transfer to the excluded families or higher orders. Appendix D gives the denominators, kinematics, and target construction. Each request has an explicit outcome; a model prediction for an unsupported native input is recorded without calling it a successful reduction.

6.1 Reserved results and loop coverage

The corrected shared model completes 43 matched nontrivial requests from the 110-request grid. The coverage record contains 21 failed reference calculations, 10 failed enlarged-region checks, 0 incomplete method comparisons, and 36 inputs above the native coordinate limit. Identity-only batches number 0. The matched outputs pass 5,976 coefficient comparisons. Table 4 retains every loop order and outcome.

Complete supplied-basis time, summed over request medians, is 63.450 seconds for native choices, 60.094 seconds for the common fixed setting, 57.125 seconds for the shared model, and 59.198 seconds for the preceding application. The learned-to-native ratio is 0.900, with repeat interval [0.867,0.919]. The learned-to-fixed ratio is 0.951, with interval [0.916,0.971]. Relative to the preceding application, the ratio is 0.965, with interval [0.900,0.979]. All ratios include inference, checking, and recovery.

The 15 matched requests from excluded families or higher orders give a learned-to-native ratio of 0.939, with repeat interval [0.897,0.978], and a learned-to-fixed ratio of 0.885, with interval [0.840,0.927]. Their physical groups were excluded from both fitting and model selection.

Table 3: The shared model across physical classes. Ratios use sums of median times on each matched set. Values below one mean less complete time. A class can contain both training-domain and held-out loop orders.

Physical class	Matched	Model/native	Model/fixed
Box with bubbles	4	0.597	0.981
Four-point ladder	4	1.109	1.042
Nonplanar box	1	1.040	1.010
Box with a virtual leg	2	1.092	0.965
Pentagon	2	0.977	0.915
Massive two-point	7	0.957	0.898
Massless two-point	7	0.793	0.988
Distinct-mass vacuum	7	0.861	0.812
One-mass vacuum	9	1.140	0.983

The model changes the initial action from native bounds on 43 matched requests and from the fixed control on 17. Thus the comparison records actual action changes as well as complete

time. Reference discovery across the requested grid takes 1056.4 seconds and the independent checks take 772.2 seconds, including unsuccessful attempts. These costs precede the matched supplied-basis timings. The new fit takes 9.5 seconds and reuses the existing cost labels; this fitting time does not include the earlier cost of generating those labels.

Table 4: Every requested loop order in the corrected-model test. "Reference" includes failed initial or enlarged-region reductions. "Method" means an incomplete matched comparison. A dash means that no complete ratio is available. The last column compares the shared model with native choices on matched nontrivial requests.

Loops	Requests	Matched	Reference	Method	Unsupported	Ratio
1	12	12	0	0	0	1.027
2	14	13	1	0	0	0.990
3	12	10	2	0	0	0.712
4	12	6	6	0	0	0.827
5	12	2	10	0	0	1.172
6	10	0	8	0	2	—
7	8	0	4	0	4	—
8	6	0	0	0	6	—
9	6	0	0	0	6	—
10	6	0	0	0	6	—
11	6	0	0	0	6	—
12	6	0	0	0	6	—

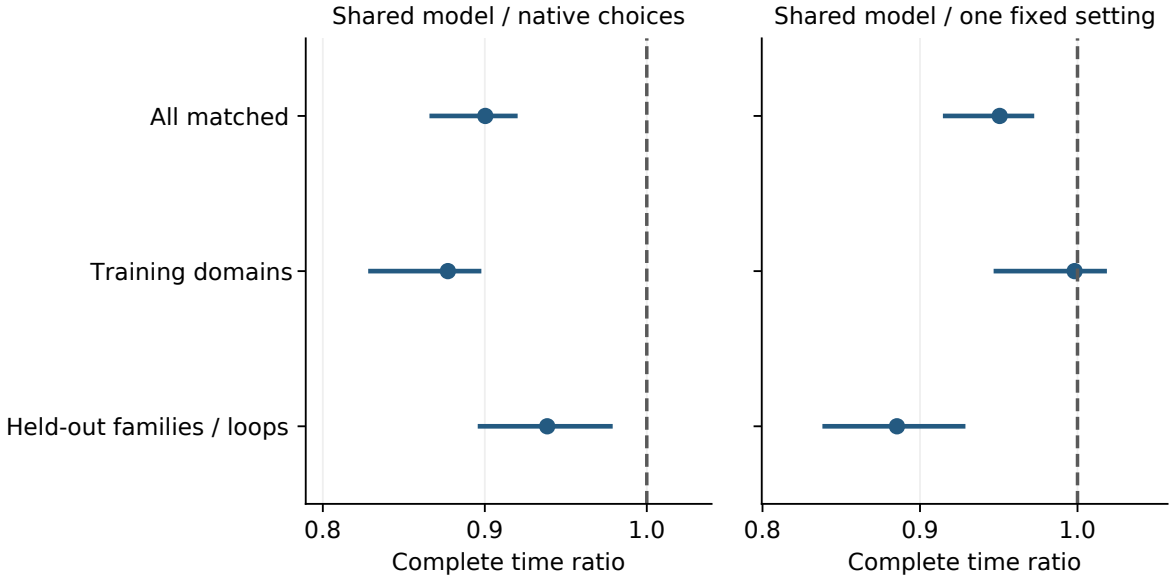


Figure 2: Complete time of the corrected shared model, separated into training domains and physical holdouts. Both controls use the same exact implementation. The fixed control has one setting for all requests. The line at one marks equal time, and the intervals describe paired repeat variation.

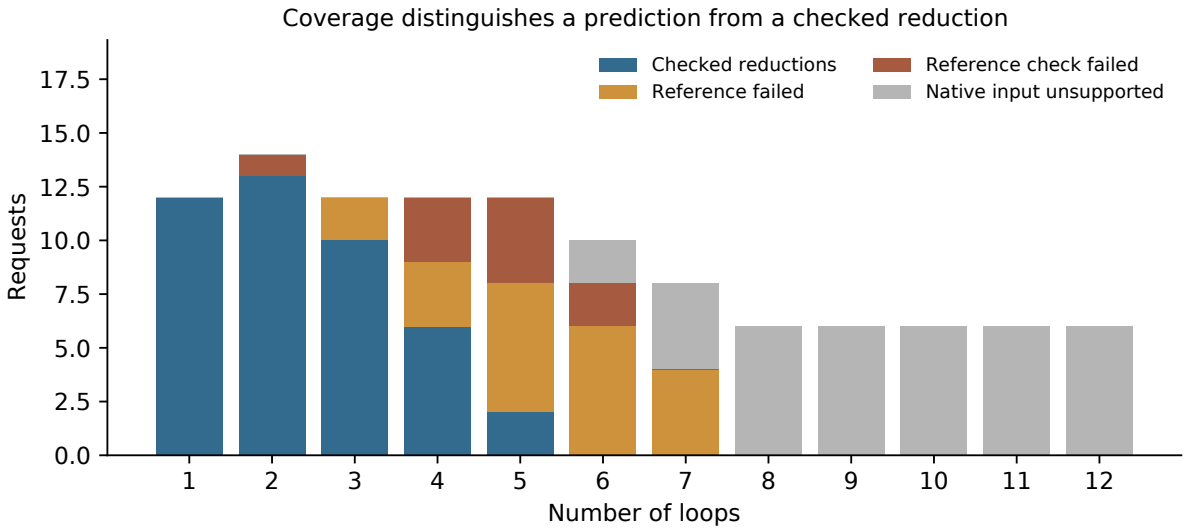


Figure 3: The loop range extends through twelve, while native completion has a smaller measured range. Predictions for unsupported inputs do not count as reductions. Failed reference checks also prevent a timing claim.

The family holdouts test transfer within this finite set of denominator structures, target powers, and rational parameter values. They do not establish performance for arbitrary graph routings, large target ranks, general symbolic kinematics, cuts, or unbounded loop order. The same physical definitions have also been used in earlier project development, although the excluded groups supply no labels or model-selection costs to this new fit. The high-loop failures and native input limits therefore remain part of the result.

7 Reducing preparation failures

The shared-model result separates two costs that require different changes. Its supplied-basis calls complete, while some initial reference calculations or enlarged-region checks fail. We therefore change the preparation workflow and train a second model on individual successful and failed attempts. This comparison includes basis lookup or discovery, both exact reductions, prediction, recovery, and process exit.

The compact workflow first looks for previously checked master indices with the same ordered propagator structure and zero-mass pattern. It reuses these indices while recomputing all coefficients at the new kinematic values. Without a stored basis, it begins with a compact seed region and treats the resulting basis as provisional. A second reduction uses that basis and a different integral ordering, and its coefficients must agree exactly. Kira's supplied-basis check rejects additional masters [5]. Agreement checks the requested reductions; it does not prove that the provisional basis has the globally smallest possible number of masters. Incomplete attempts expand the seed region, and duplicate attempts are removed.

The new model takes the same 44 structural quantities as the neural model, together with the three seed bounds, the truncation flag and level, the presence and size of a supplied basis, and the integral-ordering setting. Its 52 inputs predict log attempt time and a failure score. We fit random forests and extremely randomized forests [10, 11] and select the action with the smallest sum of predicted time and a validation-selected failure penalty. Appendix F defines the outputs and objective. The failure score ranks actions; it is not a calibrated probability.

Training uses 1,405 earlier attempts, including 389 failures, and 58 separate validation requests select the model. The selected random forest has 32 trees of maximum depth 6 and uses a 120-

second failure penalty. Fitting and model selection take 3.67 seconds, excluding the earlier cost of collecting labels. Training includes the initial shared-model trial, but excludes the distinct-mass vacuum family, nonplanar boxes, pentagons, and loop orders six and above. The new physical profile supplies no fitting or selection labels.

The reserved comparison contains 33 requests at one through seven loops, using the physical definitions in Appendix D with profile $h = 1237$. It compares expanded discovery and its enlarged-region check with three versions of the compact workflow: native bounds, the frozen neural model, and the cost-and-risk model. Each method receives the same 110-second request deadline and a 60-second cap per native call. Every first attempt is retained, and each initially successful method receives two further repetitions. All four methods use the same 128-bit integral-ordering labels, which Kira supports [5]; this setting changes the internal ordering representation rather than the exact coefficient arithmetic. The 31-coordinate interface limit remains.

Table 5: Preparation outcomes over the full reserved set. The first-pass timer includes successful and unsuccessful requests. Failed-attempt time counts the unsuccessful native calls within that pass. These totals describe work performed; a shorter failed request is not a successful speedup.

Method	First complete	All 3 complete	First pass (s)	Failed calls (s)
Expanded discovery	17/33	16/33	1041.6	684.2
Compact, no ML	23/33	23/33	757.8	573.1
Compact, neural model	23/33	23/33	869.0	663.4
Compact, cost and risk	23/33	23/33	895.1	677.6

Table 6: Preparation completions by loop order. Each entry requires all three repetitions. The compact columns use the same reuse and checking procedure with the stated decision rule. These results are separate from the earlier supplied-basis coverage in Table 4.

Loops	Requests	Expanded	Compact native	Neural	Cost and risk
1	4	4	4	4	4
2	7	6	7	7	7
3	4	3	3	3	3
4	6	2	3	3	3
5	6	1	4	4	4
6	4	0	2	2	2
7	2	0	0	0	0

Table 7: Complete preparation time on matched requests. Every included method completes all three repeats in the same master-index basis with exact coefficient agreement. The compact/native comparison includes reuse and the changed check; the two ML comparisons isolate prediction within that workflow.

Comparison	Matched	Time ratio	Repeat interval
Compact without ML / expanded	13	0.305	[0.290,0.329]
Neural model / compact without ML	22	0.982	[0.959,1.021]
Cost and risk / compact without ML	22	1.635	[1.604,1.700]
Cost and risk / neural model	23	1.536	[1.471,1.595]

The analysis performs 6,375 coefficient comparisons across the repeated and competing outputs. The reused index cache contains 24 earlier checked structures. Their retained discovery and check records cost 341.3 seconds before this experiment; failed attempts during the earlier search add further cost. The compact timings therefore describe use of an existing cache, not cost-free master discovery. In the earlier supplied-basis experiment, 129 neural-model native attempts contain 0 failures. This audit locates the larger failure cost in preparation.

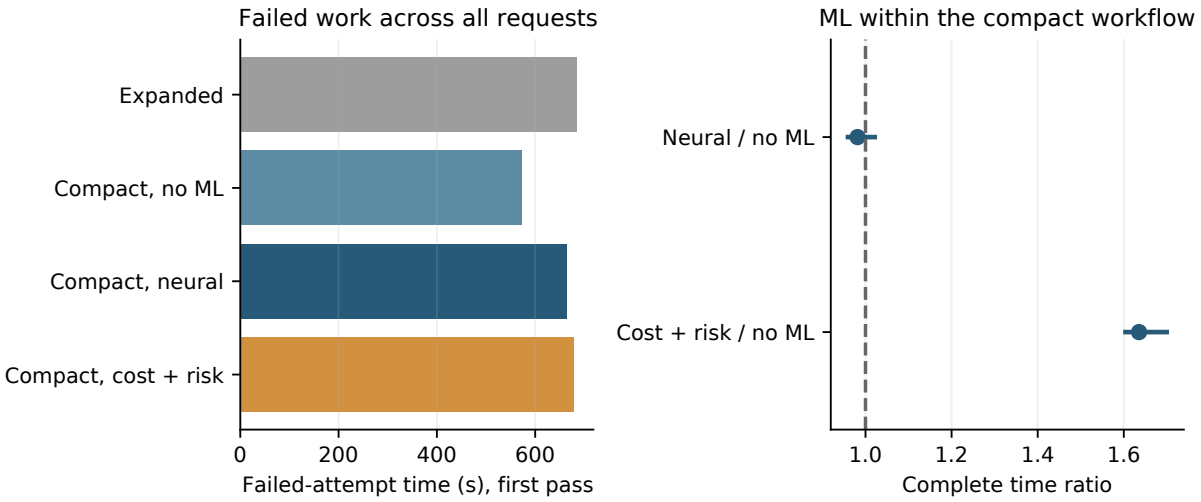


Figure 4: The preparation comparison keeps failed work and the added ML effect separate. The left panel includes failed calls from every first request. The right panel uses only exact matched completions and compares both models with the same compact workflow without ML. The dashed line marks equal time.

Compact preparation without ML compared with expanded discovery gives ratio 0.305 on 13 matched requests, with repeat interval [0.290,0.329]. The neural model within compact preparation gives ratio 0.982 on 22 matched requests, with repeat interval [0.959,1.021]. The cost-and-risk model within compact preparation gives ratio 1.635 on 22 matched requests, with repeat interval [1.604,1.700]. The cost-and-risk model increases time in this broad comparison. The neural preparation result also remains consistent with equal time. We keep the failure model and its records for analysis, while the shared neural model remains the validated supplied-basis default.

The full application first checks whether the restricted scalar massless formula applies. This is a physical condition, independent of measured time. Among the remaining requests, the neural preparation model gives ratio 0.985 on 21 matched completions, with repeat interval [0.963,1.025], relative to compact preparation without ML. These measurements do not meet both conditions for retaining learned preparation: a time interval below one and completion of every request completed by its native control. The default therefore uses native choices for preparation and retains the shared neural model for supplied-basis reduction, where the separate test establishes a gain. This deployment choice follows the candidate comparisons. The reported ratios belong to the frozen methods; they do not provide a new independent timing test of the resulting selector.

7.1 Recovery from an incomplete stored basis

The first compact trial exposed a target-dependent basis mismatch: a basis checked for earlier targets can omit masters needed by a new batch. The revised application discards a rejected cached basis after expanded seeding also fails, rediscovers without that constraint, and checks the result. It also permits an expanded target check when smaller checks fail. All steps remain inside the same 110-second request budget. A focused follow-up uses six fresh requests at two through five loops and keeps the models and index cache fixed. All three repetitions complete on 6/6 requests with ML disabled, 6/6 with the neural model, and 6/6 with the cost-and-risk model. The neural model gives a ratio of 0.783 on 6 matched requests, with repeat interval [0.743,0.797], relative to the revised workflow without ML. The cost-and-risk model gives a ratio of 0.801 on 6 matched requests, with repeat interval [0.772,0.857], relative to the revised workflow without ML. The comparison passes 2,352 coefficient checks. It tests the recovery change on fresh parameters of selected structures and does not replace the broad trial.

8 Exact loop blocks and learned costs through thirteen loops

The next extension changes the calculation around the learned choice. When the active propagators and numerators divide the loop momenta into independent blocks, we reduce each distinct block once and keep the product of its master expansions in factored form. Shared external momenta do not prevent this separation. An active numerator that joins two loop sets prevents their separate reduction. Appendix G derives the factorization and gives exact tadpole, one-mass vacuum, and massless two-point formulas used before native Kira. These routes use mathematical identities and make no learned choice.

The remaining blocks use the same 44 structural inputs and ten seed actions as the shared model. We fit eleven new candidates on all 111 available historical training requests, with 76 separate validation requests. They comprise six forests, two regularized linear regressions, and three neural networks. The forests follow the random-forest and extremely randomized-tree methods [10, 11]. Each forest averages its tree predictions. The cost labels sum the native attempts and their recovery cost, since a checked stored basis permits a single production reduction. An unsuccessful full action receives a penalty. The first selection retains the preceding network with two 48-component hidden layers. After adding the two exact convolution formulas, we compare the fitted candidates again on the historical validation requests that still need native Kira, and retune the fixed control on the same requests. This final selection precedes the corrected reserved test. The notes give the candidate settings and selection rules.

A stored basis contains integral indices, and Kira recomputes coefficients at the current parameters. During development, combining several alternative preferred master lists allowed different seed actions to return different bases. Their coefficients could not be compared entry by entry. We therefore reduce all stored preferred indices at separate development parameters and retain one checked master set per structure. A production result must use this set; otherwise the application retries with less restrictive seeds. Unknown structures require master discovery and a second exact calculation. These preparation costs remain outside the supplied-cache timing and are recorded separately in the notes.

The first reserved grid contains 79 coupled-family requests and 52 independent-block requests through thirteen loops. It records every failure and native input limit. After the basis correction, we use new parameter profiles for the coupled and product tests. Each product uses one- or two-loop blocks; a total of thirteen loops therefore tests the exact composition and its local learned decisions. It does not test a general irreducible thirteen-loop matrix. Further sets check the derived one-mass vacuum and massless two-point formulas with scalar targets and one quadratic numerator. The notes give the initial grid and all outcomes, while Table 8 gives the corrected application's paired results.

All three corrected methods share the block construction, exact formulas, checked basis, and recovery limits. The native method uses native seed bounds. The fixed method uses the preselected common action. The learned method predicts an action from the block structure. Each timing starts a fresh process and includes prediction, native equation generation, reduction, recovery, output, and exact comparison with an independent reference. Three repetitions rotate method order. Reference preparation is separate. A time ratio uses only requests for which both methods complete all three exact checks; incomplete requests remain in the coverage counts. Analytic-only requests contribute no evidence for ML.

The final validation comparison contains 58 requests that need native Kira and 18 requests handled by the exact formulas. It selects a 48-tree extremely randomized forest with maximum depth 10, with decision margin 0.0. The model weights and decision rule remain fixed throughout the corrected reserved tests.

The corrected grid contains 92 requests. The learned route completes all three exact checks on 91 of them, with 0 first-run completions that lack matching reference coefficients. Across all

methods, the experiment records 892 timed attempts and 38012 coefficient comparisons.

Table 8: Corrected application: paired time ratios on completed requests. The numerator is the learned application time; the controls share all exact changes. Times are sums of request medians in seconds. Each row has its own matched set. Analytic rows execute the same formula code and contain no ML inference. These are measurements of this study.

Comparison	Requests	Control	Learned	Ratio
Coupled families / native	18	27.400	22.715	0.829
Coupled families / fixed	18	22.471	22.715	1.011
Independent blocks / native	52	187.116	189.802	1.014
Independent blocks / fixed	52	191.693	189.802	0.990
Vacuum formula / native	11	12.441	11.300	0.908
Vacuum formula / fixed	11	11.735	11.300	0.963
Massless formula / native	10	10.895	10.796	0.991
Massless formula / fixed	10	10.866	10.796	0.994

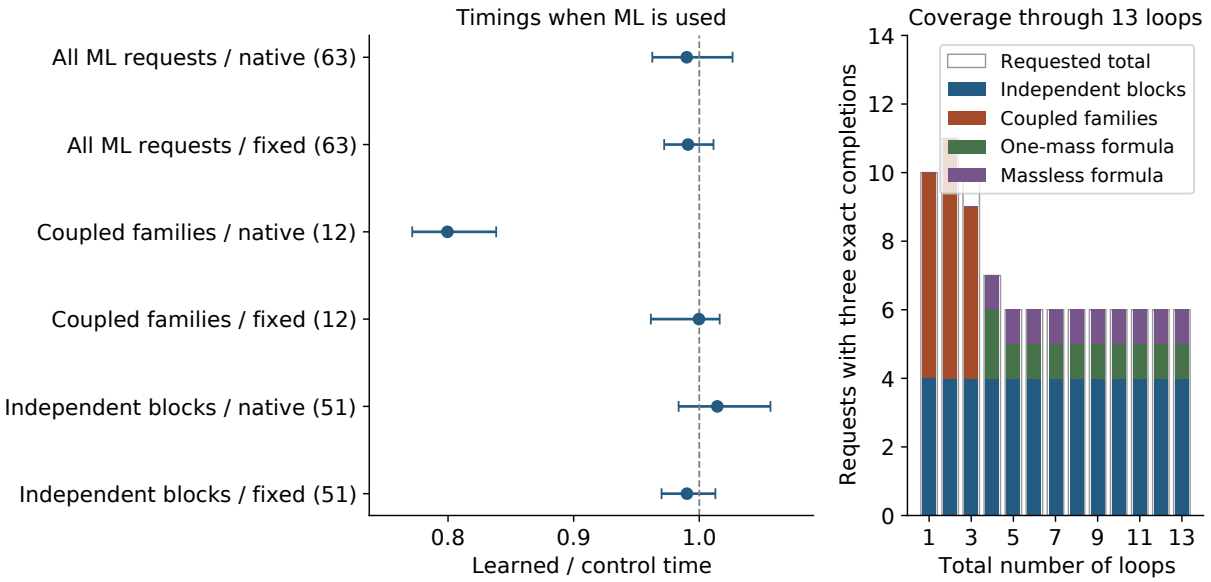


Figure 5: Left: only requests that invoke ML are included, and the controls share all exact changes. The comparisons with fixed choices include equal time within repetition variation. Intervals describe the three repetitions. Counts in parentheses identify matched requests. Right: completion depends on the integral structure. High-loop products contain small independent blocks, while each analytic formula applies to its specified denominator structure. The bar outline gives the requested total. These sets do not establish general coupled high-loop coverage.

On requests that actually invoke ML, the time ratio is 0.990, with repetition interval $[0.963, 1.026]$, on 63 matched requests. Against the fixed action on its matched ML requests, the time ratio is 0.991, with repetition interval $[0.972, 1.011]$, on 63 matched requests. The comparison with native choices does not resolve a saving beyond repetition variation. The comparison with fixed choices does not resolve a saving beyond repetition variation. These controls isolate the learned seed decision after the exact changes.

The learned method records 45 unsuccessful native attempts among 288 native calls in the timed tests. A completed request includes its failed attempts and recovery in the measured cost. Unverified learned attempts consume 101.255 seconds in addition to the paired completed-request totals. Their outcomes remain in the coverage counts.

Across all timed attempts, the learned application uses 811.364 seconds, native choices use 790.074 seconds, and the fixed choice uses 818.950 seconds. This total includes the unfinished

request and all repetitions. The learned application therefore does not reduce total time against native choices on this bounded workload.

In the 63 matched ML requests against the fixed control, the model changes the first native action in 34 requests. The remaining requests retain the fixed action and still pay the prediction cost.

Compared with the initial block application on the matched fresh coupled set, the time ratio is 1.046, with repetition interval $[0.989, 1.085]$, on 16 matched requests. On the separate one-mass vacuum targets with a completed initial block reduction, the time ratio is 0.187, with repetition interval $[0.168, 0.214]$, on 4 matched requests. For the corresponding massless numerator comparison, the time ratio is 0.781, with repetition interval $[0.724, 0.891]$, on 1 matched requests. These changes include exact formulas and revised basis handling, so they do not measure ML alone.

The separate gamma-formula checks cover 14 vacuum requests through thirteen loops and 50 target coefficients. Native Kira confirms 5 of the attempted low-loop requests, with 14 coefficient comparisons; 0 attempted native references do not complete. Higher-loop agreement between the finite-product and gamma expressions checks the derivation and implementation, without establishing a native high-loop timing gain.

The massless numerator checks add 12 requests and 60 gamma comparisons through thirteen loops. Native Kira verifies 3 low-loop requests and 15 coefficients; 0 native references do not complete. These formulas precede ML and provide no evidence for a learned gain.

9 Choosing baseline seed bounds by complete time

The exact routes in Section 8 leave a native Kira problem whenever no supported formula applies. We compare nine seed settings for this remaining work. The candidates include native bounds, bounds enlarged to contain the supplied masters, and subsector truncations. Two candidates use the least enlargement that admits the required masters under Kira's additional cutoff [5]; Appendix I derives this condition. We fit cost-sensitive trees from 31 training requests and use 20 separate requests for model selection, with all failed attempts and recovery included in the cost.

Validation retains a fixed truncation with an increased numerator-rank bound because the forest's added saving does not reach the declared deployment threshold. Thus this stage supplies an ML-free baseline for the combined method below. The notes preserve its complete model search, measurements, and earlier conservative comparison.

The reserved baseline test requests 65 reductions and verifies 64 in all three repetitions. On matched completions, the selected application takes 209.217 seconds against 213.807 seconds for original native bounds, giving a ratio of 0.979. Its ratio to the preceding application is 0.915; that comparison also includes the revised output handling. The first pass across all requests, including unfinished work, takes 280.860 seconds. The grid spans loop orders one through thirteen through specified formulas and independent products, while coupled native requests reach three loops. The three-loop ladder remains unverified within the resource limits. These are fixed-workflow results and do not establish an added gain from the fitted forest.

10 Combining learned and exact reduction methods

The complete-time comparison above retains a fixed setting for the broad request collection. The earlier sparse-seed model has a separate tested use for nonplanar two-loop integrals with higher numerator powers. We therefore combine these methods and measure their complete costs on fresh inputs. Exact formulas handle their supported families first, and independent loop blocks are reduced separately. For the remaining work, the application can use the preceding setting, a fixed sparse boundary, the trained sparse-seed expert, or the original native bounds. Every route

retains exact Kira algebra, the supplied master-basis check, and recovery after an unsuccessful attempt.

10.1 The sparse expert and the complete-time selector

The sparse expert acts before Kira generates equations. It uses the target numerator powers to select the boundary of a seed region, as defined in Section 4 and Appendix B. Its validated input domain contains one target in the specified nonplanar massless two-loop box, with seven unit physical powers and numerator rank two, three, or four. We check the ordered denominator and momentum structure before using this expert. Other blocks use the preceding fixed setting within its training ranges and return to original native bounds outside those ranges. The range check also covers numerator ranks above the broad training collection. This domain restriction permits a useful local model without assigning it an unsupported task.

The expert retains its previously trained 32-tree model with maximum depth 2. Its 12 original training requests and 10 model-selection requests precede this experiment. The model predicts log-time ratios for candidate sparse boundaries from 68 inputs: 27 family and target summaries, 40 positive- and negative-power coordinates, and the largest number of negative entries. Tree evaluation uses the saved thresholds and mean leaf values, following the earlier extremely randomized forest construction [11]. Appendix J gives the candidate boundaries and the fresh target batches. The current comparison includes feature calculation, prediction, seed enumeration, input preparation, and every recovery attempt.

We then fit an outer cost model from new complete reduction times. It receives the 51 structural and master-basis inputs of Appendix H, together with the two auxiliary numerator powers for this ordered family. Its four actions are the four routes listed above. The training cost includes the inner expert's prediction when that route uses it. We use the direct cost split of Eq. (39), so the outer model learns which complete route takes less time.

The new collection has 16 training requests and 8 validation requests with separate rational kinematics. Each training action is measured once and each validation action twice. We compare one tree or sixteen trees, maximum depths one through five, and minimum leaf sizes two, four, or seven. Validation first compares the outer prediction with the best constant route. It then compares the resulting application with the best ML-free candidate. Each deployment comparison requires at least 1.5% less validation cost and no increase in unverified requests. The fixed sparse candidate retains the boundary selected in the earlier expert study. Current validation compares that candidate with the preceding setting and original native bounds to select the ML-free control. The selected control is a fixed sparse-seed boundary. Validation enables the outer 1-tree model, with maximum depth 2 and minimum leaf size 4. The selected validation cost is 52.826 seconds, compared with 55.917 seconds for the ML-free control. These model-selection results precede the reserved test.

10.2 Fresh complete-workflow comparison

The reserved comparison contains twenty-four nonplanar target requests and ten compatibility requests. The nonplanar requests cover every distribution of numerator ranks two through four across the two auxiliary coordinates at two new kinematic profiles. Eight of these requests use index vectors absent from the new outer-model training and validation collection. The earlier inner expert has a separate training history, which includes some of these added shapes. The ten compatibility requests include native coupled families through three loops, independent products through thirteen loops, and a thirteen-loop analytic case. We freeze the expert weights, outer model, domain checks, and decisions before testing. Each request uses three repetitions with rotated method order on one worker. The preceding application runs through its actual entry point. The fixed control disables both learned predictions, while all routes retain exact coefficient

checks and the same native resource limits.

The integrated application verifies 34 of 34 requests in all three repetitions, and invokes learned prediction on 24. For 34 matched requests, the integrated method takes 156.290 seconds and the preceding complete-time application takes 161.592 seconds. This gives 3.3% less time, with ratio 0.967 and repetition interval $[0.950, 1.003]$. For 24 matched requests, the integrated method takes 136.956 seconds and native seed bounds takes 225.338 seconds. This gives 39.2% less time, with ratio 0.608 and repetition interval $[0.584, 0.640]$. For 24 matched requests, the integrated method takes 136.956 seconds and the validation-selected ML-free control takes 143.079 seconds. This gives 4.3% less time, with ratio 0.957 and repetition interval $[0.925, 0.998]$. The comparison with the selected ML-free control resolves an added saving from learned decisions beyond the measured repetition variation. For 8 matched requests, the integrated method takes 52.876 seconds and the validation-selected ML-free control takes 51.685 seconds. This gives 2.3% more time, with ratio 1.023 and repetition interval $[0.974, 1.109]$. For 8 matched requests, the integrated method takes 52.876 seconds and the preceding complete-time application takes 48.131 seconds. This gives 9.9% more time, with ratio 1.099 and repetition interval $[1.066, 1.194]$. These added index patterns expose a remaining transfer limit for the outer selector.

Table 9: Complete hybrid comparison. The first-pass time includes one attempt at every requested calculation. Ratios use sums of request medians on matched exact completions. The fixed setting is selected from the three declared ML-free candidates on validation data.

Method	Verified	First pass (s)	Integrated/method	Failed calls
Preceding application	34/34	162.721	0.967	0
Native bounds	34/34	243.584	0.636	0
Selected ML-free control	34/34	165.651	0.961	48
Integrated method	34/34	159.543	1.000	0

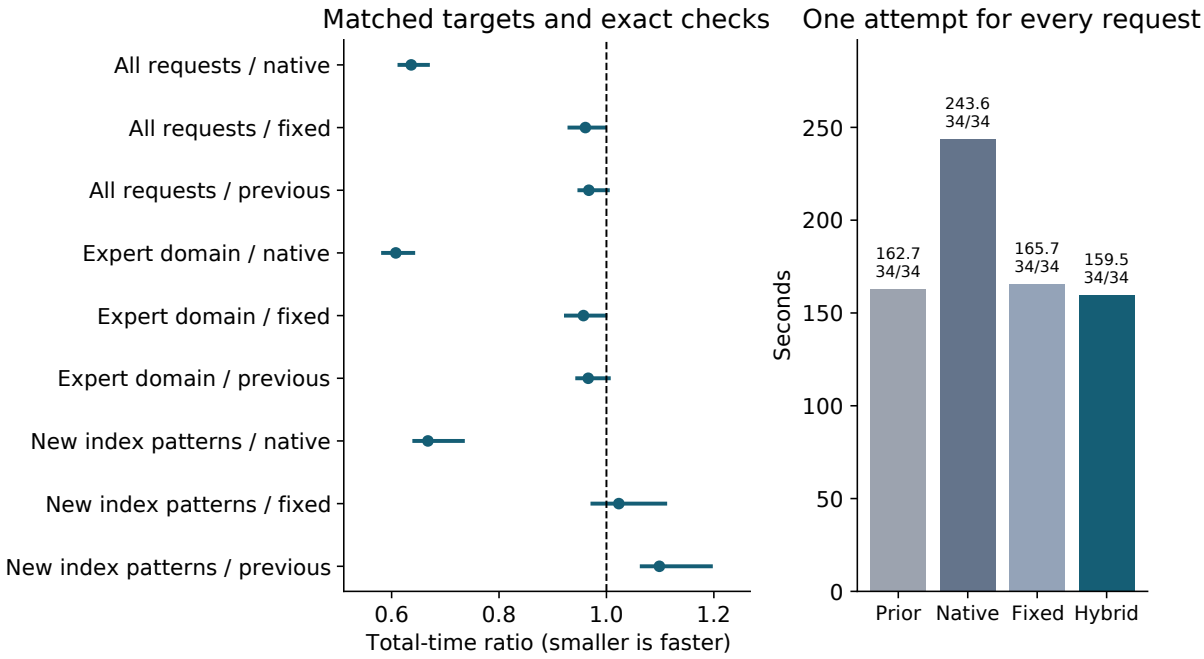


Figure 6: The left panel gives complete-time ratios for all matched requests, the sparse-expert domain, and the additional index patterns absent from outer-model training. The fixed control isolates the added learned decision. The line at one marks equal time, and horizontal bars show paired repetition variation. The right panel includes one attempt for every request and labels verified coverage. Prior and Hybrid denote the preceding and integrated applications.

Across all recorded attempts and repetitions, the integrated method takes 476.353 seconds, compared with 493.287 seconds for the ML-free setting and 487.815 seconds for the preceding application. The 128 new training and validation measurements take 1033.449 timed seconds; outer-model fitting takes 12.241 seconds. Independent reference preparation for the reserved requests takes 908.971 seconds outside the application timers. The earlier expert training and checked master-basis preparation are additional prior costs.

The comparison measures use with an existing checked basis and includes model loading, prediction, equation generation, algebra, exact checks, and recovery. The direct learned result applies to the specified nonplanar family and numerator range. The compatibility checks verify the retained routes on their stated families; the broader baseline comparison above remains a separate experiment. The application keeps native row and equation choices. These finite tests establish no globally fastest reduction or coverage of arbitrary high-loop scattering.

11 Overall interpretation

The current hybrid comparison in Section 10 tests seed settings after exact formula and independent-block reduction. Its fixed control isolates the added learned choice under the same exact checks and recovery rules.

The direct local comparisons establish savings for seed and equation decisions within their measured domains (Table 2). The shared-model comparison tests the additional question of transfer without a family-specific rule (Section 6.1). These are separate workloads and controls, so their ratios do not form a cumulative speedup. Learned row selection remains omitted because its measured gain is not resolved from timing variation. The current hybrid uses its validated sparse expert and the exact reductions in Section 8. The shared model and equation filter retain their separately tested applications, and their gains are not added to the hybrid result.

The preceding complete application combined validated learned choices with exact preparation changes. On 41 matched batches it took 56.461 seconds against 66.045 seconds for native choices, a ratio of 0.855. Its isolated new learned choice changed the seed action in only one batch. We retain that earlier experiment in the notes because its combined gain does not establish a general learned policy. Exact sector-storage changes, scaling reuse, and analytic scalar recurrences likewise remain distinct from learned gains. In particular, the scalar recurrence checks through twelve loops verify a restricted exact formula, not a high-loop ML reduction.

12 Conclusion

The integrated method applies exact formulas when their assumptions hold and reduces each distinct independent block once. It then selects a native setting or a trained sparse-seed expert for the remaining algebra. We train the outer choice from complete elapsed costs and compare it with ML-free controls. Validation retains a fixed basis-compatible setting for the broad collection and separately decides whether the sparse expert helps on its supported numerator domain.

For 34 matched requests, the integrated method takes 156.290 seconds and the preceding complete-time application takes 161.592 seconds. This gives 3.3% less time, with ratio 0.967 and repetition interval $[0.950, 1.003]$. For 24 matched requests, the integrated method takes 136.956 seconds and the validation-selected ML-free control takes 143.079 seconds. This gives 4.3% less time, with ratio 0.957 and repetition interval $[0.925, 0.998]$. The comparison with the selected ML-free control resolves an added saving from learned decisions beyond the measured repetition variation. On the 8 requests with index vectors absent from outer-model training, the method takes 9.9% more time than the preceding application.

The integrated test completes 34 of 34 requests in all three repetitions. Its first pass, including

every requested outcome, takes 159.543 seconds. The broader baseline study completes 64 of 65 requests and retains the unfinished three-loop ladder in its cost and coverage records.

The learned result remains specific to the stated nonplanar family and numerator powers. Exact formulas and independent products extend the checked loop range under their mathematical assumptions. The workflow retains native row and equation choices, while the earlier equation-selection gain remains a separate measured result. Master-basis preparation and model training precede the supplied-basis timers. These measurements establish the supported benefits and remaining failures without asserting a globally fastest reduction for every integral.

A IBP identities, row operations, and exactness

The starting total-derivative identity follows from integration by parts in a region where the integral converges, followed by dimensional analytic continuation [3, 2]. Write $P_{\boldsymbol{\sigma}} = \prod_i D_i^{-\sigma_i}$ and $\partial_k = \partial/\partial\ell_k$. The product rule gives

$$\partial_k \cdot (v P_{\boldsymbol{\sigma}}) = (\partial_k \cdot v) P_{\boldsymbol{\sigma}} - \sum_i \sigma_i (v \cdot \partial_k D_i) D_i^{-\sigma_i - 1} \prod_{j \neq i} D_j^{-\sigma_j}. \tag{4}$$

For $v = \ell_h$, the first factor is $d\delta_{kh}$; for an external momentum it is zero. Completeness of the inverse-propagator coordinates permits us to write $v \cdot \partial_k D_i = u_i + \sum_j v_{ij} D_j$, where u_i and v_{ij} are coefficients fixed by the propagators and kinematics, with the invariant and mass terms included in u_i. Integrating Eq. (4) therefore gives

$$0 = (\partial_k \cdot v) I_{\boldsymbol{\sigma}} - \sum_i \sigma_i u_i I_{\boldsymbol{\sigma} + \boldsymbol{e}_i} - \sum_{i,j} \sigma_i v_{ij} I_{\boldsymbol{\sigma} + \boldsymbol{e}_i - \boldsymbol{e}_j}. \tag{5}$$

The unit vector $\boldsymbol{e}_i$ raises index i by one. This derivation explains how a seed and an operator generate a row of the reduction system.

Let q be a source row and k a pivot column with $A_{qk} \neq 0$. Solving the identity $\sum_j A_{qj} I_j = 0$ for I_k gives $I_k = -\sum_{j \neq k} (A_{qj}/A_{qk}) I_j$. Subtracting A_{rk}/A_{qk} times that zero identity from row r yields

$$A'_{rj} = A_{rj} - \frac{A_{rk}}{A_{qk}} A_{qj}, \qquad A'_{rk} = 0. \tag{6}$$

Every step preserves the represented relation. Repetition until only prescribed master columns remain gives Eq. (3). Removing seeds or equations preserves the validity of the retained identities, but can prevent this elimination from finishing. We therefore require reduction to the prescribed basis and exact agreement with the reference coefficients. Generic kinematics avoid singular pivot denominators; special kinematic limits require their own treatment.

B Sparse seeds and the equation-classifier score

Let $\mathcal{T}$ be the set of target index vectors. For seed $\boldsymbol{\sigma}$, define the extra positive power $D(\boldsymbol{\sigma}) = \sum_i \max(0, \sigma_i - 1)$ and numerator rank $R(\boldsymbol{\sigma}) = \sum_i \max(0, -\sigma_i)$. These definitions count denominator powers beyond one and powers in the numerator. Let $P = \{i : \sigma_i > 0\}$, let h be the number of absent propagators relative to the largest top sector, and let D_T, R_T be the largest target values. The sparse model uses

$$q(\boldsymbol{\sigma}, \mathcal{T}) = \min_{\boldsymbol{a} \in \mathcal{T}} \sum_{i \notin P} \max\{0, \max(0, -\sigma_i) - \max(0, -a_i)\}. \tag{7}$$

For a fixed target, project the nonnegative seed numerator power onto the interval from zero to the target power. The distance is the positive excess in that coordinate. Adding coordinate

distances and minimizing over targets gives Eq. (7). To construct a variable-width region, we choose an affine width in the sector deficit, extra powers, and rank. The integer lattice requires its floor, and a maximum with zero prevents a negative width. Applying the same construction to rank and dot allowances gives the restrictions, with fitted parameters $\theta_0, \ldots, \theta_6$,

$$\begin{aligned} q &\le \max\{0, \lfloor \theta_0 + \theta_1 h + \theta_2 D + \theta_6 R \rfloor\}, \\ R &\le \max\{0, \lfloor R_T + \theta_3 - \theta_4 h \rfloor\}, \qquad D \le \max\{0, \lfloor D_T + \theta_5 \rfloor\}. \end{aligned} \tag{8}$$

These inequalities define the search family, and each selected region still requires a sufficiency check. Native outer bounds, retained corners, and admissible target and master seeds also apply. The tested targets have $D_T = 0$, and $\theta_5 = 1$ remains fixed. The notes give the finite search ranges and fitted vectors.

For B training requests, the boundary search minimizes $J = B^{-1} \sum_b \log[T_b(\boldsymbol{\theta})/T_b^{(0)}]$, where T_b includes reduction, checking, and recovery. Exponentiating gives the geometric mean of the time ratios, which derives the equal-request objective. Failed bounded reductions receive a 300-second training penalty. The subsequent cost predictor compares learned and fixed boundaries before the reserved timing experiment; these training penalties never become successful test times.

B.1 Equation inputs and sampling

The equation classifier uses seventeen inputs. Four describe the family and request: $L, E, N, \log(1+|\mathcal{T}|)$. Three describe the seed: positive-index count, extra positive power, and numerator rank. Two subtract the batch's maximum extra power and rank from the corresponding seed quantities. Three describe the seed sector: whether it contains a target, and the differences from its target power and rank maxima. Two describe proximity: exact target membership and the minimum sum of absolute index differences. The latter is the defined distance $\min_{\boldsymbol{a} \in \mathcal{T}} \sum_i |\sigma_i - a_i|$. The operator index and the two overall target maxima complete the vector. These are definitions of the inputs, all available before coefficient construction.

Let N_{cy} be the native number of records of class $y \in \{0, 1\}$ in batch c. The total number of records is $N_c = N_{c0} + N_{c1}$. We sample $n_{cy} = \min(2000, N_{cy})$ records from each class without replacement. Every sampled record receives weight $w_{cy} = N_{cy}/(n_{cy} N_c)$. The total weight of that class is $n_{cy} w_{cy} = N_{cy}/N_c$, so the two classes together give unit batch weight. For a leaf with weighted class totals W_0, W_1, the score is

$$p = \frac{W_1}{W_0 + W_1}. \tag{9}$$

This weighted class fraction is the classifier's score, without a claim of probability calibration. The tree chooses splits using weighted Gini impurity [9]. Expanding $W[1 - p^2 - (1-p)^2]$, with $W = W_0 + W_1$, gives $2W_0 W_1/W$. A threshold retains an equation if $p \ge \tau$; the whole-seed variant retains every operator if the maximum score over that seed's operators reaches τ. A maximum reaches a threshold exactly when at least one element reaches it, so this variant retains a superset of the source classifier's selected equations. It still requires the sufficiency check above.

C Shared structural model, actions, and fitting

Let n be the number of physical propagators and E the number of independent external momenta. Let B_{ih} be one when loop momentum ℓ_h appears in propagator momentum q_i, and zero otherwise. Table 10 groups the 27 base inputs. Means run over the target batch unless another set is specified. A weighted incidence multiplies the relevant numerator power or extra denominator power by the corresponding row sum of B before summing over propagators.

Table 10: The base structural inputs. Each quantity is computed from the family definition and target indices before Kira generates equations.

Group	Quantities	Number
Family dimensions	L, E, N, n; number of symbolic invariants	5
Propagator structure	Massive-entry count; coupled-physical-entry count; maximum physical row and column sums of B; mean external-momentum incidence over all entries	5
Target batch	Log of one plus target count; number of target sectors	2
Power ranges	Minimum, mean, and maximum extra positive powers and numerator rank	6
Positive indices	Minimum, mean, and maximum number of positive indices	3
Further index limits	Largest absolute index; maximum positive-power sum; maximum number of absent physical denominators	3
Weighted powers	Mean dots on massive entries; mean numerator-weighted and dot-weighted loop incidence	3

We add eight ratios: N/n, n/L, massive-entry fraction, coupled-physical-entry fraction, maximum dots divided by n, maximum rank divided by $N - n$, maximum physical loop incidence divided by L, and maximum physical loop degree divided by n. Each denominator is replaced by at least one. Six further features are the mean fractions of indices equal to $-2, -1, 0, 1, 2, 3$. The final three are $\log[1 + L(L + E)]$, $\log(1 + n)(1 + D_T)$, and $\log[1 + \max(1, N - n)](1 + R_T)$. These definitions give 44 features. Permuting loop names or entries within the physical and auxiliary groups preserves the pooled counts when the indices are permuted consistently. A change of momentum routing need not preserve them, and identical features need not identify identical reduction problems.

Let (r_0, s_0, d_0) be the target maxima of positive-power sum, numerator rank, and extra positive power. Table 11 defines ten candidate actions. Bounds are nonnegative and r is at least the number t_{top} of positive entries in the largest top sector. Kira's subsector restriction uses [5]

$$s \le \max\{1, t - l + 1, s_{\max,\text{sector}}\}, \tag{10}$$

where t is that sector's positive-index count, l is the chosen parameter, and the last term is an explicitly assigned sector limit.

Table 11: The shared model's candidate seed configurations. Changes are relative to the target maxima. A dash disables the additional subsector restriction.

Action	Δr	Δs	Δd_0	l
Native	0	0	0	—
Numerator expansion	0	1	0	—
Tight	−1	−1	−1	—
Reduced dots	−1	0	−1	—
Reduced numerator rank	0	−1	0	—
Expanded	1	1	1	—
Truncated native	0	0	0	$t_{\text{top}} - s_0$
Truncated expansion	0	1	0	$t_{\text{top}} - s_0$
Truncated reduced dots	−1	0	−1	$t_{\text{top}} - s_0$
Relaxed truncation	0	0	0	$t_{\text{top}} - s_0 - 1$

For request c and action a, define the cost label $y_{ca} = \log(T_{ca}/T_{c0})$, with native action 0. Because the denominator is positive and independent of a, and the logarithm is increasing, minimizing the unmodified label selects the action with the least recorded time. Training clips this label to $[-4, 5]$ after assigning failures at least 180 seconds or three times the measured attempt

cost, whichever is larger. Clipping limits extreme labels and can merge their order; the validation comparison therefore uses the recorded costs directly.

The neural regression model uses composed affine maps and componentwise activations [12]. The selected network standardizes each feature by its training mean and standard deviation, with a minimum scale of 10^{-6}. With standardized vector z, its outputs are defined by

$$h_1 = \mathrm{ReLU}(W_1^{\mathsf{T}} z + b_1), \quad h_2 = \mathrm{ReLU}(W_2^{\mathsf{T}} h_1 + b_2), \quad \widehat{y} = W_3^{\mathsf{T}} h_2 + b_3, \tag{11}$$

where $\mathrm{ReLU}(u) = \max(0, u)$ acts on each component. Both hidden vectors have 48 components and the output has ten components, one per action. We fit squared prediction error with a squared-weight penalty of strength 0.1 using the regression implementation of Ref. [12]. For n_{tr} training requests, the corresponding objective is

$$\mathcal{L} = \frac{1}{2n_{\mathrm{tr}}} \sum_{c,a} (\widehat{y}_{ca} - y_{ca})^2 + \frac{0.1}{2n_{\mathrm{tr}}} \sum_{j} \|W_j\|_F^2. \tag{12}$$

To motivate the first term, assign a formal residual $e_{ca} = \widehat{y}_{ca} - y_{ca}$ the density $p(e) = (2\pi)^{-1/2} \exp(-e^2/2)$. Then

$$-\frac{1}{n_{\mathrm{tr}}} \log \prod_{c,a} p(e_{ca}) = \frac{1}{2n_{\mathrm{tr}}} \sum_{c,a} e_{ca}^2 + \frac{10}{2} \log(2\pi).$$

Removing the constant and adding the stated quadratic penalty gives Eq. (12). This construction motivates a regression objective; the study does not infer a Gaussian distribution of timing errors.

The candidate set also contains trees, random forests, and extremely randomized forests at depths 2, 4, and 7, with at least three samples per leaf; forests contain 48 trees [9, 10, 11]. Two ridge fits use regularization strengths 1 and 10, and the neural candidates have two hidden layers of either 24 or 48 components. Neural fitting uses L-BFGS with at most 500 iterations [12]. The initial implementation exposed an inconsistent scaling between the installed multi-output loss and its gradient. We corrected the loss to match the summed-output gradient, checked it by finite differences, and refitted using the same training and validation inputs. The initial comparison remains in the notes; the corrected model uses a new reserved profile. Candidate convergence warnings remain recorded, and no fit certifies a global optimum.

For every candidate, five-fold cross-validation withholds whole ordered graph structures, with feature scaling fitted within each training fold. The separate validation set selects a single fixed action. We compare that action and native bounds as global anchors, with margins $0, 0.05, 0.12, 0.25$. If b is the anchor and $a_* = \arg\min_a \widehat{y}_a$, the policy changes to a_* only when $\widehat{y}_{a_*} < \widehat{y}_b - \eta$. Exponentiation gives the predicted ratio condition $\exp(\widehat{y}_{a_*} - \widehat{y}_b) < \exp(-\eta)$. We select the candidate minimizing the worse of its grouped-fold and separate-validation ratios to native cost. This rule is fixed before the reserved test. A score below 0.97 permits default deployment; it is evidence from validation, not a guarantee for future inputs.

The selected candidate is the 48-component network, with action 7 from Table 11 (numbered from zero) as anchor and margin 0.05. Training uses only historical training records after the physical holdouts. Their historical timers differ in whether initial imports are included; all new reserved controls use the same complete fresh-process timer. The training record contains every source and model candidate. A failed new seed attempt retries native and expanded bounds with duplicate configurations removed. The shared model never uses target coefficients or measured test time as an input.

D Physical inputs and the loop-order test

The following equations define the physical inputs for our benchmark families. All denominators follow Eq. (1). Let $Q = \sum_{i=1}^{L} \ell_i$. For $L \geq 2$, the vacuum families contain

$$q_i = \ell_i \quad (1 \leq i \leq L), \qquad q_{L+1} = Q. \tag{13}$$

The one-mass case gives only the final entry a nonzero mass; the massive case assigns distinct nonzero masses to all physical entries. The one-loop vacuum is a single massive tadpole. The two-point families replace the final momentum by $p - Q$, with either all zero masses or distinct nonzero masses. Their single independent external invariant is p^2.

For the four-point families, choose massless external momenta with $p_1 + p_2 + p_3 + p_4 = 0$, and write $P = p_1 + p_2$, $U = p_1 + p_2 + p_3$, $s = P^2$, $t = (p_2 + p_3)^2$. Expanding these squares gives $2p_1 \cdot p_2 = s$, $2p_2 \cdot p_3 = t$, and $2p_1 \cdot p_3 = -s - t$. The physical momenta of the L-loop ladder are

$$\{\ell_i, \ell_i - P : 1 \le i \le L\} \cup \{\ell_i - \ell_{i+1} : 1 \le i < L\} \cup \{\ell_1 - p_1, \ell_L - U\}. \tag{14}$$

We order the loop momenta first, the P-shifted momenta next, then adjacent differences, and finally the two end entries. The box with bubble subgraphs has

$$(q_1, q_2, q_3, q_4) = (\ell_1, \ell_1 - p_1, \ell_1 - P, \ell_1 - U), \qquad (q_{2j+1}, q_{2j+2}) = (\ell_j, \ell_j - \ell_1), \quad 2 \le j \le L. \tag{15}$$

All these four-point propagators are massless. This box family has bubble subgraphs and is a restricted connected scattering class.

The additional massless nonplanar box uses the routing of Ref. [1]:

$$\begin{aligned}(q_1, \ldots, q_9) = (&\ell_1, \ell_2, p_1 - \ell_1, p_2 - \ell_2, p_1 + p_2 - p_3 - \ell_1, \\ &p_1 - \ell_1 + \ell_2, p_1 - p_3 - \ell_1 + \ell_2, \ell_1 - p_3, \ell_2 - p_1 - p_2).\end{aligned} \tag{16}$$

Its first seven entries are physical. Here the outgoing-momentum convention gives $p_i^2 = (p_1 + p_2 - p_3)^2 = 0$, $s = (p_1 + p_2)^2$, and $t = (p_1 - p_3)^2$. Expansion gives $2p_1 \cdot p_3 = -t$ and $2p_2 \cdot p_3 = s + t$. The pentagon uses five massless external momenta with zero sum and physical momenta ℓ, $\ell - p_1$, $\ell - p_1 - p_2$, $\ell - p_1 - p_2 - p_3$, and $\ell - p_1 - p_2 - p_3 - p_4$. Adjacent invariants are $s_{ij} = (p_i + p_j)^2$. Squaring complementary momentum sums gives, for example, $2p_1 \cdot p_3 = s_{45} - s_{12} - s_{23}$. The box with one virtual external leg uses $(\ell, \ell - p_1, \ell - p_1 - p_2, \ell - p_3)$, with $p_i^2 = 0$ for $i = 1, 2, 3$ and $v = (p_1 + p_2 - p_3)^2$; its remaining product is $2p_2 \cdot p_3 = s + t - v$.

There are L loop squares, $L(L-1)/2$ products of distinct loop momenta, and LE loop–external products. Adding these counts gives the complete coordinate number

$$N = \frac{L(L+1)}{2} + LE. \tag{17}$$

This is the representation used by Kira [5]. We append squared momentum differences when their scalar-product coefficient vectors increase the exact rank, until all N coordinates are independent. The candidate order follows loop momenta and then external momenta. Auxiliary indices are nonpositive. The tested interface admits at most 31 coordinates; this is an implementation limit, not an upper bound on the mathematics of IBP. It permits the defined vacuum inputs through seven loops, two-point inputs through six, and four-point inputs through five. Completion remains a separate test.

For the final fresh profile $h = 1213$, choose $m^2_{\text{one}} = 2h{+}17$, $m_i^2 = 2h{+}19{+}6(i{-}1)$, $p^2 = -2h{-}29$, $s = -1009{-}12h$, and $t = -1021{-}14h$, using only the variables required by each of the six recursive families. All values use a common squared-momentum unit. Additional scattering profiles follow the explicit assignments in Table 12. The independent reference increases the target's positive-power, numerator, and dot limits by one; the check increases the numerator allowance once more and changes integral ordering. Every checked reference and every failure is retained.

Let $a_i^{(0)} = 1$ on the n physical entries and zero otherwise. For $j = 0, \ldots, \min(n, 3) - 1$, set $i_j = 1 + [(h + j) \bmod n]$ and, when $N > n$, $k_j = n + 1 + [(h + j) \bmod (N - n)]$. The dotted batch uses $\boldsymbol{a}^{(0)} + 2\boldsymbol{e}_{i_j}$; the mixed batch uses $\boldsymbol{a}^{(0)} + \boldsymbol{e}_{i_j} - \boldsymbol{e}_{k_j}$. Without an auxiliary entry, the latter has only the extra physical power. The three additional structures use the earlier generator, which puts a second physical dot on the next entry when no auxiliary entry exists. The notes give its

Table 12: Additional scattering kinematics at profile $h = 1201$ for the initial shared-model trial and $h = 1213$ for the corrected-model test. These expressions define the chosen inputs in a common squared-momentum unit.

Family	Invariants
Nonplanar box	$s = -113 - 18h$, $t = -127 - 10h$
Box with virtual leg	Same s, t; $v = -149 - 12h$
Pentagon	$(s_{12}, s_{23}, s_{34}, s_{45}, s_{51}) = (-11, -13, -17, -19, -23) - 20h(1, 1, 1, 1, 1)$

full historical target definitions. These are indexed Feynman integrals whose reductions come from exact Kira calculations.

For the retained two-loop sparse-seed test, the planar box has physical momenta $(\ell_1, \ell_1 - p_1, \ell_1 - p_1 - p_2, \ell_2, \ell_2 - p_1 - p_2, \ell_2 - p_1 - p_2 - p_3, \ell_1 - \ell_2)$ and auxiliary momenta $(\ell_1 - p_3, \ell_2 - p_1)$. Its convention is $2p_1 \cdot p_2 = s$, $2p_1 \cdot p_3 = t$, and $2p_2 \cdot p_3 = -s - t$. The nonplanar family is Eq. (16). Both use unit physical powers and numerator ranks two through four on one or two auxiliary entries. The equation-transfer comparison also includes the three-loop vacuum with momenta $(\ell_1, \ell_2, \ell_3, \ell_1 - \ell_2, \ell_2 - \ell_3, \ell_1 - \ell_3)$ and the massive two-point ladder $(\ell_1, \ell_1 - p, \ell_2, \ell_2 - p, \ell_3, \ell_3 - p, \ell_1 - \ell_2, \ell_2 - \ell_3, \ell_1 - \ell_3)$, whose final entry is auxiliary. For the three-loop vacuum the squared masses are $(1, 1, 1, 13, 13, 53 + 2h)$; for the two-point ladder they are $(1, 0, 0, 0, 0, 0, 0, 0)$ on the physical entries, with $p^2 = -1009 - 12h$. The earlier sparse-seed fit uses profiles $h = 831, 832$ for training, $841, 842$ for validation, and $851, 852$ for the 20-request test. Its planar invariants are $(-1009 - 12h, -1021 - 14h)$ and its nonplanar invariants are $(-113 - 18h, -127 - 10h)$. A single-direction numerator carries rank two, three, or four; a distributed numerator splits rank three or four into its ceiling and floor halves on the two auxiliary entries. The equation-filter training uses profiles $511, 512$, validation uses 513, and its new-index comparison uses $605, 606$. The notes preserve the complete finite target enumeration and source records for those earlier experiments.

E Timing ratios and uncertainty

Let T_{cmj} be the elapsed time for request c, method m, and repetition j. For the matched set $\mathcal{C}$, define

$$S_m = \sum_{c \in \mathcal{C}} \operatorname{median}_j T_{cmj}, \qquad R_{m/b} = \frac{S_m}{S_b}, \qquad \text{time saving} = 100(1 - R_{m/b})\%. \tag{18}$$

The sum describes that fixed workload. Subtracting S_m from the baseline total S_b and dividing by S_b gives the saving formula; its speedup factor is $S_b/S_m = 1/R_{m/b}$. A matched request must complete the compared methods and repetitions with exact agreement. We also report all excluded requests, so these ratios are conditional on completion.

For the new test, we resample the three paired repetition indices within each request 2,000 times, recompute its medians and the ratio of totals, and take the 2.5 and 97.5 percentiles. This applies the bootstrap principle [13] to the measured repetition pairs. The interval describes variation on this fixed workload, with limited resolution from three repeats. It does not give a confidence guarantee for unseen integral families or arbitrary loop order. Training, reference discovery, and unsuccessful comparisons are reported separately from matched supplied-basis times.

F Learning attempt cost and failure risk

For recorded attempt i, let t_i be its elapsed time in seconds and f_i be one if it fails and zero otherwise. We define the two regression targets as $u_i = \log \max(0.001, t_i)$ and $v_i = 3f_i$. A tree

leaf predicts the mean of each target; averaging the trees gives predictions $\widehat{u}$ and $\widehat{v}$ [10]. The factor three gives the failure output more weight in the two-output squared-error split objective. For n attempts, that objective is proportional to

$$\sum_{i=1}^{n}\left[(\widehat{u}_i - u_i)^2 + (\widehat{v}_i - 3f_i)^2\right]. \tag{19}$$

To obtain the leaf means, differentiate each sum of squared residuals with respect to its constant leaf prediction. The derivatives are $2\sum_i(\widehat{u} - u_i)$ and $2\sum_i(\widehat{v} - 3f_i)$; setting them to zero gives the two sample means. Equation (19) is the chosen fitting objective, without an assumption that every failure has the same physical cause.

The runtime score for candidate action a is defined as

$$C_a = \exp[\mathrm{clip}(\widehat{u}_a, -5, 8)] + \lambda\,\mathrm{clip}(\widehat{v}_a/3, 0, 1). \tag{20}$$

Here $\mathrm{clip}(z, l, h) = \min(h, \max(l, z))$ and λ has units of seconds. The exponential reverses the log-time target, and division by three reverses the failure-label scaling. Their sum defines a time-based ranking with a failure penalty; it is not a calibrated expected-time estimate. Without a supplied basis, candidate actions exclude the three settings that reduce dot or numerator bounds below the native target limits. All ten settings are available when a basis is supplied.

We compare 32-tree random and extremely randomized forests at depths four and six, with at least five training attempts per leaf [10, 11]. Each candidate is evaluated with $\lambda \in \{0, 20, 60, 120\}$ seconds. For each validation request, it chooses one recorded action. Model selection minimizes

$$V = \frac{\sum_c T_{c,a(c)} + 60\sum_c f_{c,a(c)}}{\sum_c T_{c,0}}, \tag{21}$$

where T is the earlier complete action cost including its recovery, f records failure of that action's first attempt, and action zero uses native bounds. Unsuccessful complete validation records receive a 180-second penalty. The denominator is independent of the candidate, so minimizing V minimizes the penalized numerator. The extra 60 seconds deliberately emphasizes avoiding a failed first action even when recovery already contributes to T. The model, penalty, and basis cache are fixed before the reserved comparison.

The new grid contains one target shape per selected family and loop order: dotted targets at odd orders and mixed targets at even orders, as defined in Appendix D. The one-mass vacuum uses orders 1, 2, 4, 5, 6, 7; the distinct-mass vacuum uses 2, 4, 5, 6, 7; each two-point class uses 2 through 6; the ladder uses 1 through 5; and the box with bubbles uses 2 through 5. The nonplanar box, pentagon, and box with one virtual leg add three requests. The cache uses only the master indices from the earlier profile 1201. Stored coefficients never enter prediction or the new reduction. Its structure key distinguishes the ordered momenta and zero versus nonzero mass and invariant assignments. A matching key is only a proposed basis: the new exact calculations must establish target reduction at the new parameters.

Every initially successful method receives three measurements. A paired time ratio includes only requests that complete all three repeats for both methods, use the same master-index set, and agree coefficient by coefficient. Repetition and cross-method checks use exact rational simplification. Different provisional bases, incomplete repetitions, and failed requests remain in the outcome records without a matched ratio. This makes the time comparison conditional on equal completed work.

The focused recovery comparison uses profile $h = 1249$ for the distinct-mass vacuums at four and five loops, the massless and massive three-loop two-point families, the three-loop box with bubbles, and the two-loop ladder. It uses dotted targets at odd orders and mixed targets at even orders. Three repetitions compare the same compact workflow with native bounds, the neural model, and the cost-and-risk model. Model weights and cache entries are unchanged. The revised recovery rule follows the basis rejection observed in the preceding experiment; this follow-up therefore tests that specific correction.

G Independent loop blocks and exact convolution formulas

G.1 When a product reduction is exact

Suppose the loop momenta partition into q disjoint sets $B_1, \dots, B_q$ and every factor with a nonzero exponent depends on loop momenta from one set. External momenta may be shared. Equation (1) then has the form

$$I_{\boldsymbol{a}} = \int \prod_{b=1}^{q} d\mu_b \, f_b = \prod_{b=1}^{q} \int d\mu_b \, f_b = \prod_{b=1}^{q} I^{(b)}_{\boldsymbol{a}_b}. \tag{22}$$

Here $d\mu_b$ contains the normalized loop measures in block b, and $\boldsymbol{a}_b$ selects its active indices. The equality follows first in a convergence domain by iterated integration and then by dimensional analytic continuation, with the convention of Ref. [2]. A numerator that contains momenta from two proposed blocks prevents this factorization and joins those blocks in our construction. An unused auxiliary coordinate does not join them.

If block b reduces as $I^{(b)}_{\boldsymbol{a}_b} = \sum_j r^{(b)}_j M^{(b)}_j$, where $r^{(b)}_j$ are its coefficients and $M^{(b)}_j$ are its masters, substitution into Eq. (22) gives

$$I_{\boldsymbol{a}} = \prod_b \left(\sum_j r^{(b)}_j M^{(b)}_j \right). \tag{23}$$

This expression retains the complete reduction without distributing the product. Distribution would contain up to $\prod_b n_b$ terms if block b has n_b terms, since each expanded term chooses one entry from each factor. The factored expression stores $\sum_b n_b$ entries. These counts follow from the product construction; they do not claim that an arbitrary coupled integral can be factorized. Identical blocks with identical indices and kinematics require one reduction per request.

G.2 A massive tadpole

Let $T_a = \int d^dk/(i\pi^{d/2})(k^2 - m^2 + i0)^{-a}$. The total derivative of $k^\mu (k^2 - m^2)^{-a}$ gives

$$0 = dT_a - 2a(T_a + m^2 T_{a+1}), \qquad T_{a+1} = \frac{d - 2a}{2am^2} T_a. \tag{24}$$

We used $k^2 = (k^2 - m^2) + m^2$ in the first equality. Repeated substitution reduces each positive integer power to T_1. For $m^2 = 0$, the integral has no scale and vanishes in dimensional regularization [2]. This route makes no learned decision.

G.3 One massive line and massless vacuum lines

For $L \geq 2$, define $Q = \sum_{i=1}^{L} k_i$ and

$$V_{\boldsymbol{a},b} = \int \prod_{i=1}^{L} \frac{d^dk_i}{i\pi^{d/2}} \frac{1}{\prod_i (k_i^2 + i0)^{a_i} (Q^2 - m^2 + i0)^b}. \tag{25}$$

The scalar formula below assumes positive integer a_i, b, nonzero m^2, and no external momenta. It applies to this denominator structure at any L; our checks stop at thirteen loops.

To obtain its index dependence, work first in Euclidean signature. We use analytic continuation in d and the indices before evaluating the integer cases. Define $\Gamma(z) = \int_0^\infty u^{z-1} e^{-u} du$ for $\Re z > 0$, with analytic continuation elsewhere. The Schwinger identity and a Gaussian integral give

$$\begin{aligned} (k_E^2)^{-a} &= \frac{1}{\Gamma(a)} \int_0^\infty dt \, t^{a-1} e^{-tk_E^2}, \\ \int \frac{d^dk_E}{\pi^{d/2}} \frac{e^{2ik_E \cdot x}}{(k_E^2)^a} &= \frac{1}{\Gamma(a)} \int_0^\infty dt \, t^{a-1-d/2} e^{-x^2/t} = \frac{\Gamma(d/2 - a)}{\Gamma(a)} (x^2)^{a-d/2}. \end{aligned} \tag{26}$$

The final equality uses $u = x^2/t$ and the defining gamma integral. These parameter and continuation conventions agree with Ref. [2]. Put $A_0 = \sum_i a_i$ and $\alpha = A_0 - (L-1)d/2$. Multiplying the L transforms and inverting them gives the massless convolution proportional to

$$(Q_E^2)^{-\alpha} \frac{\Gamma(\alpha)}{\Gamma(d/2-\alpha)} \prod_i \frac{\Gamma(d/2-a_i)}{\Gamma(a_i)}. \tag{27}$$

The transform normalization is independent of the indices and cancels in the ratios used below. The remaining radial integral is

$$\begin{aligned} \int \frac{d^d Q_E}{\pi^{d/2}} \frac{(Q_E^2)^{-\alpha}}{(Q_E^2+m^2)^b} &= \frac{(m^2)^{d/2-\alpha-b}}{\Gamma(d/2)} \int_0^\infty \frac{u^{d/2-\alpha-1}}{(1+u)^b}\, du \\ &= (m^2)^{d/2-\alpha-b} \frac{\Gamma(d/2-\alpha)\Gamma(\alpha+b-d/2)}{\Gamma(d/2)\Gamma(b)}. \end{aligned} \tag{28}$$

The beta integral follows by $v = u/(1+u)$. Combining Eqs. (27) and (28) cancels $\Gamma(d/2-\alpha)$. Wick rotation supplies the relative sign $(-1)^{A_0+b-(L+1)}$ in our Minkowski denominator convention. Let $P = L+1$, $\Delta = A_0 + b - P$, and $(z)_n = \Gamma(z+n)/\Gamma(z)$. Dividing by the all-one integral gives

$$\frac{V_{\boldsymbol{a},b}}{V_{\mathbf{1},1}} = \frac{(-1)^\Delta (m^2)^{-\Delta} (P - Ld/2)_\Delta \, [L-(L-1)d/2]_{A_0-L}}{(b-1)! \prod_i [(a_i-1)!\,(d/2-a_i)_{a_i-1}]}. \tag{29}$$

Thus the production formula uses finite products, while the independent algebraic check uses the gamma ratios before expansion. If a shifted massless power is zero, its momentum can be integrated after shifting to Q, leaving a scaleless factor. If $b = 0$, the remaining massless integrals are also scaleless. These terms vanish under the stated assumptions.

G.4 One quadratic numerator

The tested extension allows one numerator v^2, where $v = \sum_i c_i k_i$ with fixed coefficients c_i. We write $V[v^2]$ for Eq. (25) with v^2 inserted in the numerator. Define $\beta_i = a_i - d/2$, $\lambda = \sum_i \beta_i$, and

$$C_v = \left(\sum_i c_i \beta_i \right)^2 - \sum_i c_i^2 \beta_i^2 = 2 \sum_{i<j} c_i c_j \beta_i \beta_j. \tag{30}$$

This equality is the expansion of the square. In the Fourier representation, a momentum insertion differentiates its corresponding factor in Eq. (26). The cross terms give $-C_v (x^2)^{\lambda-1}$, while multiplication of the full convolution by Q_E^2 gives

$$-\frac{1}{4} \partial_x^2 (x^2)^\lambda = -\lambda(\lambda + d/2 - 1)(x^2)^{\lambda-1}. \tag{31}$$

The derivative follows by applying the product rule twice. Dividing these two expressions, then returning to Minkowski signature, yields

$$V[v^2] = \sum_i c_i^2 V_{\boldsymbol{a}-\boldsymbol{e}_i,b} + \frac{C_v}{\lambda(\lambda+d/2-1)} \left(V_{\boldsymbol{a},b-1} + m^2 V_{\boldsymbol{a},b} \right). \tag{32}$$

The diagonal term uses k_i^2 to cancel one denominator power; the last term uses $Q^2 = (Q^2 - m^2) + m^2$. Equations (29) and (32) give rational coefficients in d and m^2. Apparent singularities require analytic limits. Higher numerator powers and other vacuum graphs remain outside this formula.

G.5 Massless two-point integrals with a quadratic numerator

Let $P = L + 1$, $q_i = k_i$ for $i \le L$, and $q_P = p - \sum_{i=1}^{L} k_i$. The massless two-point family is

$$W_{\boldsymbol{a}}(p) = \int \prod_{i=1}^{L} \frac{d^d k_i}{i\pi^{d/2}} \prod_{j=1}^{P} (q_j^2 + i0)^{-a_j}, \qquad p^2 \neq 0. \tag{33}$$

Here all a_j are positive integers. Applying Eq. (26) to the P massless factors and inverting their product gives the convolution coefficient

$$\mathcal{C}(\boldsymbol{a}) = \frac{\Gamma(A - Ld/2)}{\Gamma(Pd/2 - A)} \prod_{j=1}^{P} \frac{\Gamma(d/2 - a_j)}{\Gamma(a_j)}, \qquad A = \sum_j a_j. \tag{34}$$

The momentum power is $Ld/2 - A$, as follows either from the inverse transform or from mass dimension. Dividing by the all-one result and cancelling the relative Wick-rotation signs for integer $A - P$ gives

$$\frac{W_{\boldsymbol{a}}}{W_{\boldsymbol{1}}} = (p^2)^{P-A} \frac{\mathcal{C}(\boldsymbol{a})}{\mathcal{C}(\boldsymbol{1})}. \tag{35}$$

The gamma shift identity turns this ratio into the finite rising products used by the scalar implementation. The derivation uses the same dimensional continuation as the preceding vacuum calculation.

Write $W[v^2]$ for Eq. (33) with numerator v^2. For $v = \sum_{i=1}^{L} c_i k_i + c_p p$, use $p = \sum_{j=1}^{P} q_j$ to write $v = \sum_{j=1}^{P} w_j q_j$, where $w_i = c_i + c_p$ for $i \le L$ and $w_P = c_p$. Define $\beta_j = a_j - d/2$, $\lambda = \sum_j \beta_j$, and $C_w = (\sum_j w_j \beta_j)^2 - \sum_j w_j^2 \beta_j^2$. The derivative calculation in Eqs. (30) and (31) now applies to all P factors. Since their total momentum is p, it gives

$$W[v^2] = \sum_{j=1}^{P} w_j^2 W_{\boldsymbol{a}-\boldsymbol{e}_j} + \frac{C_w p^2}{\lambda(\lambda + d/2 - 1)} W_{\boldsymbol{a}}. \tag{36}$$

A term with a removed massless line has a scaleless integration and vanishes. Equations (35) and (36) reduce the supported scalar and single-numerator targets to $W_{\boldsymbol{1}}$. The formula allows an external momentum in the numerator and excludes multiple numerator factors, nonzero internal masses, and more general two-point graphs.

H Derivation of the complete-time training rule

For a target batch i and an initial Kira setting a, let T_{ia} be the measured time from application launch to exact comparison and process completion. This includes loading, selection, equation generation, algebra, and every recovery attempt. We define the training cost

$$\widetilde{T}_{ia}^{(q)} = \begin{cases} T_{ia}^{(q)}, & \text{verified}, \\ \max(220\,\mathrm{s}, T_{ia}^{(q)}), & \text{unverified}, \end{cases} \qquad C_{ia} = \operatorname{median}_q \widetilde{T}_{ia}^{(q)}. \tag{37}$$

Here q labels repetitions. The 220-second penalty is a training choice that makes an unfinished calculation costly; it does not estimate its unknown completion time. The results use measured seconds and completion counts, with no penalty substituted into reported speedups.

We construct a tree from structural features of the target integrals and their required master basis. In a leaf containing training requests S, choosing one action a incurs total cost $\sum_{i \in S} C_{ia}$. Minimizing this expression gives the leaf action and cost,

$$a_S = \arg\min_a \sum_{i \in S} C_{ia}, \qquad J(S) = \min_a \sum_{i \in S} C_{ia}. \tag{38}$$

A candidate feature threshold divides S into disjoint sets S_- and S_+. The two leaves can choose their actions separately, so their minimum combined cost is $J(S_-) + J(S_+)$. We therefore choose the threshold that gives the largest positive decrease

$$\Delta J = J(S) - J(S_-) - J(S_+). \tag{39}$$

This derivation trains the partition directly from elapsed reduction cost. Cost-sensitive algorithm selection also uses unequal penalties for expensive choices [14]. Our particular split rule follows from Eqs. (38)–(39); it does not certify a globally optimal tree or reduction path.

Each leaf stores the mean cost for every candidate action. A single tree predicts these means, while a forest averages the means from its reached leaves. If $\widehat{C}_a$ is the predicted cost, a_0 is the validation-selected anchor, δ is a decision margin, and $\tau = 0.002\,\mathrm{s}$ is the decision allowance, the candidate $a_* = \arg\min_a \widehat{C}_a$ replaces the anchor only when

$$\widehat{C}_{a_*} + \tau < (1-\delta)\widehat{C}_{a_0}. \tag{40}$$

The condition requires the predicted saving to exceed the allowance and margin. Validation must also show at least 1.5% less cost than the best fixed candidate, with no increase in unverified requests, before the application enables learned decisions. Inputs outside the recorded structural ranges use native settings. These tests control deployment; they do not guarantee a saving on a new input.

The master-basis check also informs the seed region. For an integral with indices $\boldsymbol{a}$, define $r(\boldsymbol{a}) = \sum_j \max(a_j, 0)$, $s(\boldsymbol{a}) = \sum_j \max(-a_j, 0)$, $u(\boldsymbol{a}) = \sum_j \max(a_j - 1, 0)$, and $t(\boldsymbol{a}) = \sum_j \mathbf{1}_{a_j>0}$. These are the positive-power sum, numerator rank, number of dots, and number of positive indices. For the six basis-compatible candidates, we enlarge the corresponding bounds to contain the supplied basis. The unadjusted native candidate retains its original bounds. Kira's additional subsector restriction is $s \le \max(1, t-\ell+1, s_{\mathrm{set}})$, where s_{set} is the explicitly assigned sector limit [5]. Our first selector uses the stronger sufficient condition

$$\ell \le \min_M \{t(M) - s(M)\}. \tag{41}$$

We lower a proposed cutoff when needed to satisfy Eq. (41). It gives $t(M) - \ell + 1 \ge s(M) + 1$, so each master passes the extra cutoff once the outer bounds contain it. This conservative choice can admit more seeds than necessary. It does not prove that the resulting identities suffice, so the application retains the exact master-basis check and less restrictive recovery settings.

H.1 Target batches and basis features

The denominator definitions and rational kinematic profiles follow Appendix D. We specify the new target powers here so that the training and test batches have an explicit mathematical definition. Let n be the number of physical denominators, let N include the auxiliary coordinates, and define $b_j = 1$ for $j \le n$ and $b_j = 0$ otherwise. For profile integer h and target number $q = 0, \ldots, \min(n, 3) - 1$, set

$$i_q = 1 + ((h+q) \bmod n), \qquad k_q = n + 1 + ((h+q) \bmod (N-n)), \qquad v_q = 1 + (i_q \bmod n). \tag{42}$$

The index k_q is used only when $N > n$. With $\boldsymbol{e}_j$ denoting unit vector j, the four target shapes are

$$\boldsymbol{a}_q^{\mathrm{dots}} = \boldsymbol{b} + (1 + \mathbf{1}_{q=1})\boldsymbol{e}_{i_q}, \tag{43}$$

$$\boldsymbol{a}_q^{\mathrm{mixed}} = \boldsymbol{b} + \boldsymbol{e}_{i_q} - \boldsymbol{e}_{k_q}, \tag{44}$$

$$\boldsymbol{a}_q^{\mathrm{rank2}} = \boldsymbol{b} - 2\boldsymbol{e}_{k_q}, \tag{45}$$

$$\boldsymbol{a}_q^{\mathrm{pinched}} = \boldsymbol{b} - \boldsymbol{e}_{i_q} + \boldsymbol{e}_{v_q}. \tag{46}$$

Here $\mathbf{1}_{q=1}$ equals one for $q = 1$ and zero otherwise. If there is no auxiliary coordinate, the mixed shape omits its numerator term and the rank-two shape becomes $\boldsymbol{b} + 2\boldsymbol{e}_{i_q}$. The pinched shape removes one physical denominator and increases the next power by one. Training uses the first three shapes, validation uses the first two, and testing uses all four. We remove requests already handled by the exact formulas when defining the native training and validation collection. The pentagon structure is also excluded from both sets.

Independent products use the dotted shape for each vacuum, massive two-point, or ladder block. Their nonplanar box blocks and one-loop box remainders instead use $\boldsymbol{b} + 2\boldsymbol{e}_{i_q}$ for each target. Blocks share external invariants and have disjoint loop momenta; cross-block auxiliary indices remain zero. At even loop order, the separate analytic test uses $\boldsymbol{b} + \boldsymbol{e}_{i_q} - \boldsymbol{e}_{k_q}$ when an auxiliary coordinate exists, and omits the numerator otherwise. At odd loop order it uses $\boldsymbol{b} + 2\boldsymbol{e}_{i_q}$. The notes give the disjoint profile integers and preserve every resulting index vector in the measurement records.

For a nonempty master basis $\mathcal{B}$, define $r_B = \max_{M\in\mathcal{B}} r(M)$, $s_B = \max_{M\in\mathcal{B}} s(M)$, $u_B = \max_{M\in\mathcal{B}} u(M)$, and $\ell_B = \min_{M\in\mathcal{B}}[t(M) - s(M)]$. Let s_T and u_T be the corresponding target maxima, and write $x_+ = \max(x, 0)$. The seven added features are

$$\boldsymbol{z}_B = (|\mathcal{B}|, r_B, s_B, u_B, \ell_B, (s_B - s_T)_+, (u_B - u_T)_+). \tag{47}$$

These quantities describe the required basis before reduction. They contain no coefficient values. The implementation uses zeros for the basis summaries when the list is empty and applies no basis-dependent enlargement in that case.

H.2 The seven candidate settings

Let r_T, s_T, u_T be target maxima and let t_{top} be the largest positive-index count among the declared top sectors. The unadjusted native candidate uses $(\max(t_{\text{top}}, r_T), s_T, u_T)$ and no subsector truncation. For an adjusted candidate with offsets $(\Delta r, \Delta s, \Delta u)$, the bounds are

$$(r, s, u) = (\max(t_{\text{top}}, r_T + \Delta r, r_B), \max(s_T + \Delta s, s_B), \max(u_T + \Delta u, u_B)). \tag{48}$$

Table 13 completes the action definition. A dash disables the extra subsector cutoff. The native region, raised rank, and expanded region provide recovery choices when a restricted region fails its exact basis check.

Table 13: The six basis-compatible candidates, in addition to the unadjusted native candidate.

Candidate	$(\Delta r, \Delta s, \Delta u)$	Subsector cutoff ℓ
Compatible native	$(0, 0, 0)$	–
Increased rank	$(0, 1, 0)$	–
Enlarged region	$(1, 1, 1)$	–
Truncated	$(0, 0, 0)$	$\min(t_{\text{top}} - s_T, \ell_B)$
Truncated with increased rank	$(0, 1, 0)$	$\min(t_{\text{top}} - s_T, \ell_B)$
Relaxed truncation	$(0, 0, 0)$	$\min(t_{\text{top}} - s_T - 1, \ell_B)$

I A tighter condition for the master-basis cutoff

The cost objective, target definitions, and 51 inputs remain those in Appendix H. The new model learns from fresh profiles, and all nine candidate costs for one request are measured together on the same worker. The notes give the training and test profiles and the selected model parameters.

Kira restricts the numerator rank in a sector through [5]

$$s \le \max\{1, t - \ell + 1, s_{\text{set}}\}. \tag{49}$$

Here t is the sector's positive-index count, ℓ is the truncation parameter, and s_{set} is the rank limit explicitly assigned to that sector. A master with rank $s_M \le \max(1, s_{\text{set},M})$ already satisfies this additional cutoff for any ℓ. For every other master, Eq. (49) reduces to $s_M \le t_M - \ell + 1$. Rearranging gives $\ell \le t_M + 1 - s_M$.

Let $\mathcal{H}$ contain precisely the masters with $s_M > \max(1, s_{\text{set},M})$. If $\ell_0 = t_{\text{top}} - s_T$ is the proposed truncation value, the largest value no greater than ℓ_0 that admits all required masters is therefore

$$\ell_* = \min\left(\ell_0, \min_{M \in \mathcal{H}} \{t_M + 1 - s_M\}\right). \tag{50}$$

An empty inner set imposes no bound, so $\ell_* = \ell_0$ in that case. Taking the minimum enforces each master's inequality, while any larger value would violate at least one of the active bounds or exceed the proposed value. Thus Eq. (50) is the least enlargement needed for this particular cutoff-admission condition. It does not determine a sufficient IBP system or a minimum reduction time.

The two added candidates use ℓ_* with the ordinary or increased numerator-rank bound. Their outer positive-power, numerator-rank, and dot bounds still contain the supplied master basis. The explicitly seeded top sectors use the candidate's assigned rank bound; other sectors have $s_{\text{set}} = 0$ unless explicitly declared. The previous seven candidates remain available, so the trained policy can select a conservative setting when its complete cost is lower. Every candidate keeps the exact coefficient and master-basis checks, and failed attempts retain their recovery cost in training and testing.

J Definition of the hybrid expert comparison

The sparse expert uses the nonplanar denominators in Eq. (16). Its targets have seven unit physical powers. For numerator rank $R = 2, 3, 4$, we include

$$\boldsymbol{a}_R^{(1)} = (1, 1, 1, 1, 1, 1, 1, -R, 0), \qquad \boldsymbol{a}_R^{(2)} = (1, 1, 1, 1, 1, 1, 1, 0, -R). \tag{51}$$

For $R = 3, 4$, we also include

$$\boldsymbol{a}_R^{(12)} = (1, 1, 1, 1, 1, 1, 1, -\lceil R/2 \rceil, -\lfloor R/2 \rfloor). \tag{52}$$

These definitions give eight training and validation targets per kinematic profile. The reserved collection additionally includes every vector $(1, 1, 1, 1, 1, 1, 1, -j, -(R - j))$ for $j = 0, \ldots, R$ and $R = 2, 3, 4$, giving twelve targets per profile. We use $s = -113 - 18h$ and $t = -127 - 10h$, with $h = 2301, 2303$ for training, 2309 for validation, and $2311, 2317$ for reserved tests. The additional two outer-model inputs are $\max(0, -a_8)$ and $\max(0, -a_9)$, appended to the 51 inputs in Appendix H. Their coordinate meanings remain fixed within this ordered family.

The outer actions are the preceding basis-compatible setting, the expert's fixed sparse boundary, its learned boundary, and original native bounds. The best ML-free control minimizes validation cost over all but the learned boundary. The cost-sensitive tree objective follows Eqs. (37)–(40), with complete costs for these four actions. The outer prediction is retained only if it improves on the best constant action by at least 1.5%. The resulting application must also improve on the best ML-free candidate by at least 1.5%, with no increase in unverified requests. A constant choice of the learned expert still performs that expert's target-dependent boundary prediction.

For the inner expert, the boundary coefficients $\theta_0, \ldots, \theta_6$ have the definitions in Appendix B. The saved forest predicts log-time ratios for these candidates. It changes the fixed candidate

only when the predicted decrease exceeds its frozen margin. Table 14 records the candidate coefficients; the native row applies no sparse boundary. Target and master seeds that satisfy the outer native bounds remain explicitly included. The native master-basis and coefficient checks determine acceptance, and a failed sparse attempt returns to less restrictive native work.

Table 14: Frozen sparse-boundary candidates used by the inner expert. Coefficients define the seed region of Eq. (8).

Candidate	θ_0	θ_1	θ_2	θ_3	θ_4	θ_5	θ_6
0: native	–	–	–	–	–	–	–
1	1	0	0	1	0	1	0
2	2	0	0	1	0	1	0
3	1	0	0	0	0	1	0
4	1	0	0	0	1	1	0
5	1	0	0	-1	0	1	0
6	1	0	0	1	1.5	1	0
7	1.5	-0.25	0	1.25	0	1	-0.25
8	2	0	-0.25	1	0	1	0
9	2.5	0	0	1.25	0	1	-0.25

The fixed boundary index is 4 and the inner decision margin is 0. The ten compatibility requests use profile 2321 for the three original coupled native families, a pentagon, a rank-three massive two-point target, and the one-mass analytic family, and profiles $2327 + 4L$ for four independent products. Their denominators and target rules remain those in Appendices D and H.1. They provide checks at loop orders one, two, three, four, eight, twelve, and thirteen.

For the selected tree, let $u = \max(0, -a_8)$ be the first auxiliary numerator power and let $R = \max(0, -a_8) + \max(0, -a_9)$ be the total numerator rank. The first split tests $u \leq 2.5$, and its lower branch tests $R \leq 3.5$. Since both quantities are integers, the deployed choice within the expert domain is

$$\pi(u, R) = \begin{cases} \text{learned sparse boundary}, & u \leq 2,\ R \leq 3, \\ \text{fixed sparse boundary}, & u \leq 2,\ R = 4, \\ \text{preceding basis-compatible setting}, & u \geq 3. \end{cases} \tag{53}$$

Here $\pi(u, R)$ is the selected reduction route. Each leaf selects the route with its lowest predicted complete cost, subject to the decision allowance of Eq. (40). For the first two leaves, the predicted saving relative to the anchor exceeds the 0.002-second allowance. The third leaf already selects the anchor, so the allowance leaves its choice unchanged. These comparisons give Eq. (53). Thus the outer tree can select a fixed or native setting even within the sparse expert's supported family. The exact checks and recovery rules still apply to each selected route.

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